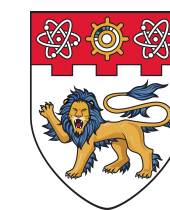


Measuring Common Phase in Ultracold Bose Gases

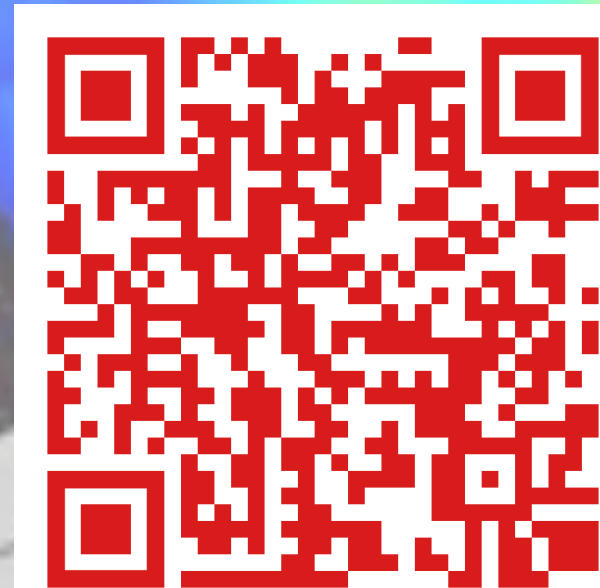
Towards Detecting Entanglement in Quantum Field Simulators

Taufiq Murtadho
RQI Circuit (Online, Jul. 2026)



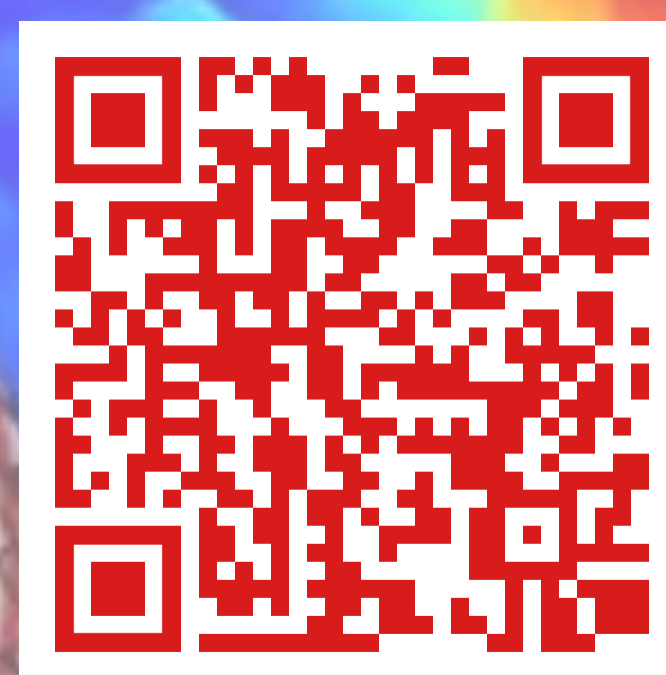
NANYANG
TECHNOLOGICAL
UNIVERSITY
SINGAPORE

$\langle N|Q^h \rangle$
NANYANG QUANTUM HUB



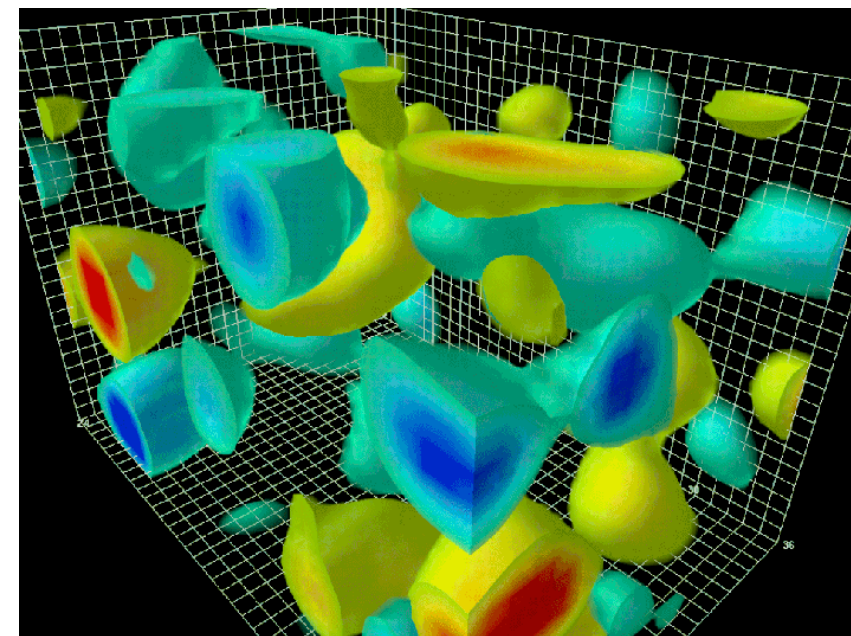
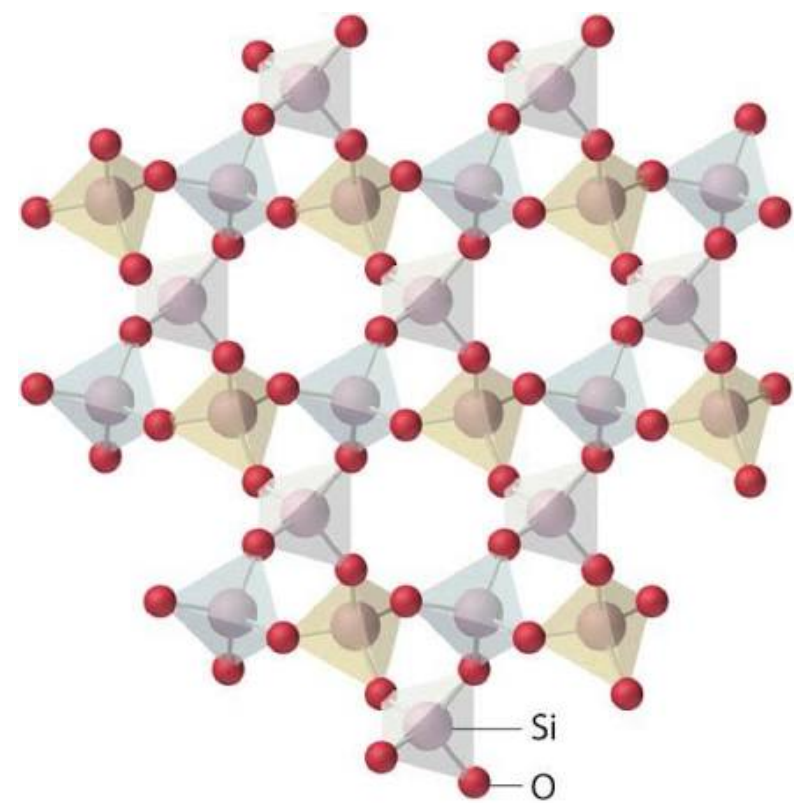
Motivation & Context

Parallel 1D Bose Gases as Quantum Field Simulator



Quantum Simulation

Credit: LibreText, Forbes

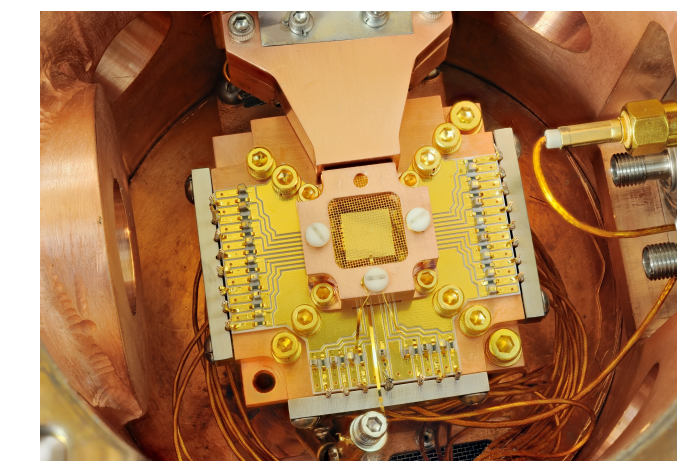
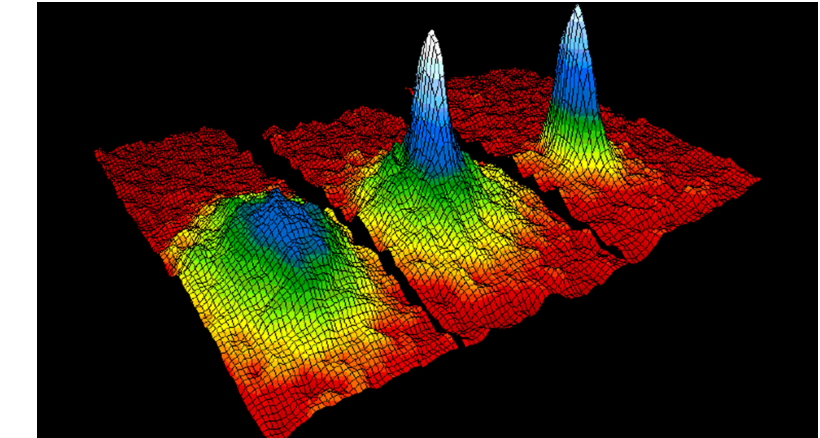
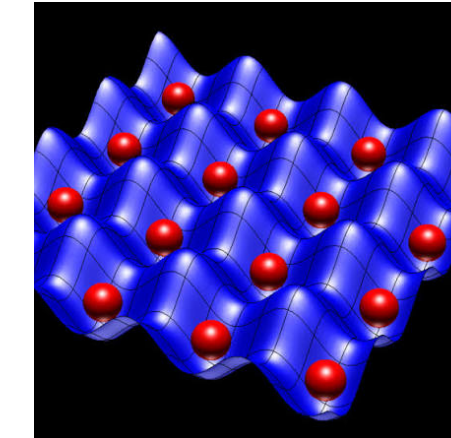


Mapping (Digital/Analog)



Insights

Credit: NIST, Wikimedia



Complex natural phenomena:
strongly correlated quantum materials,
quantum field theories (QFT),

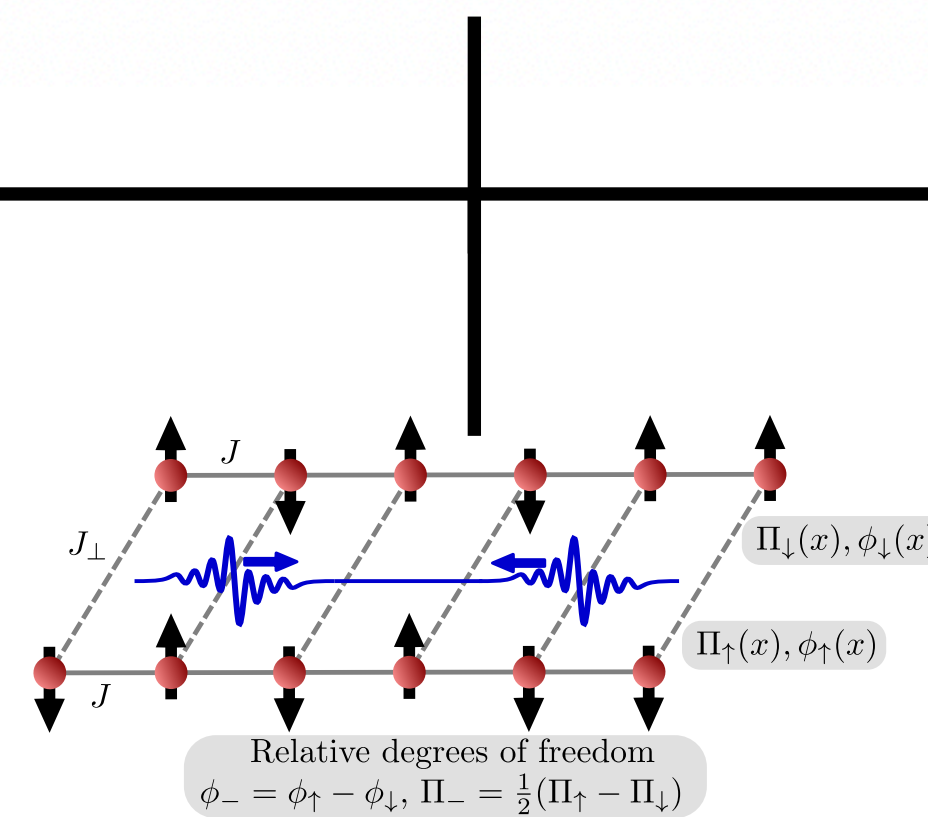
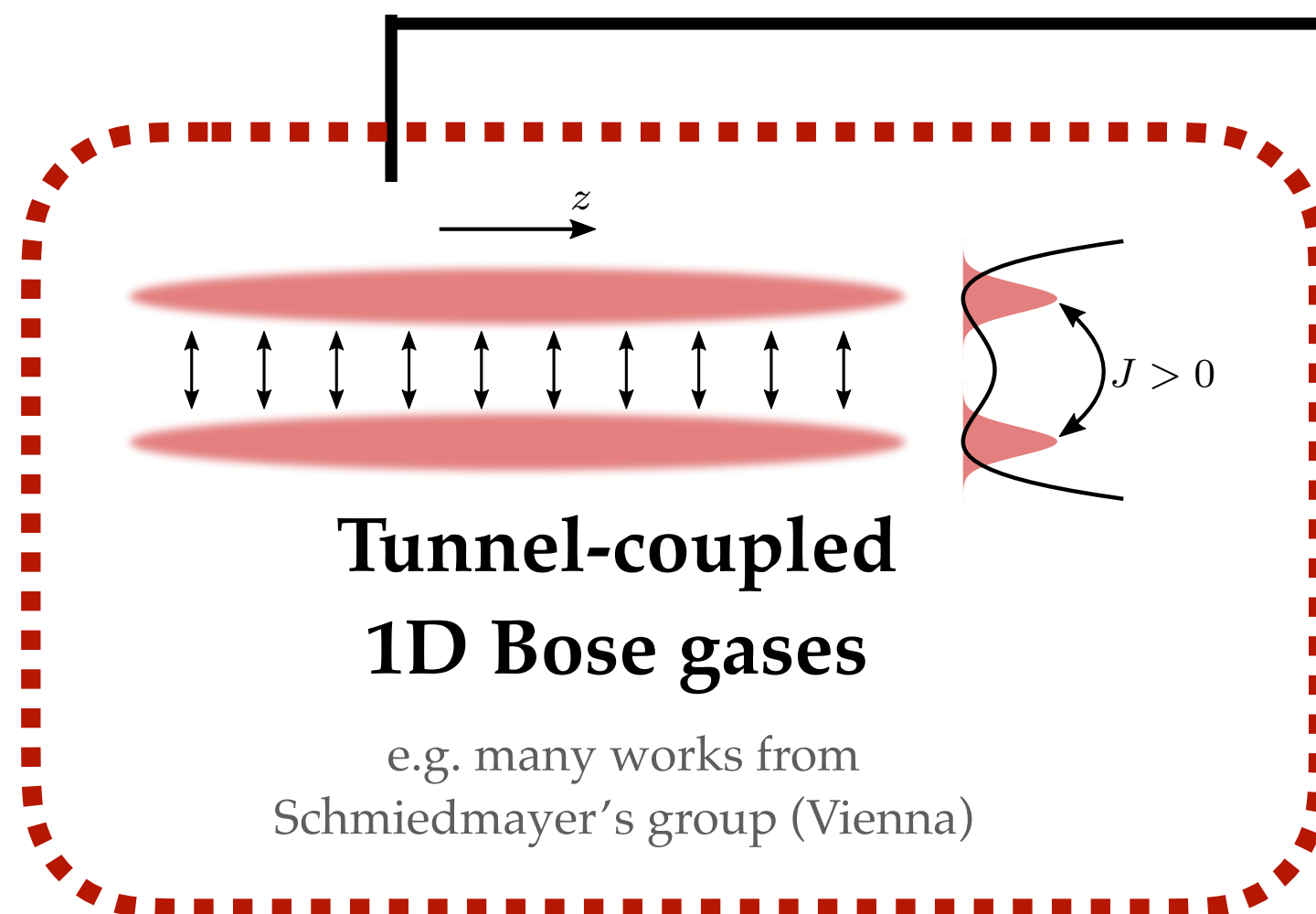
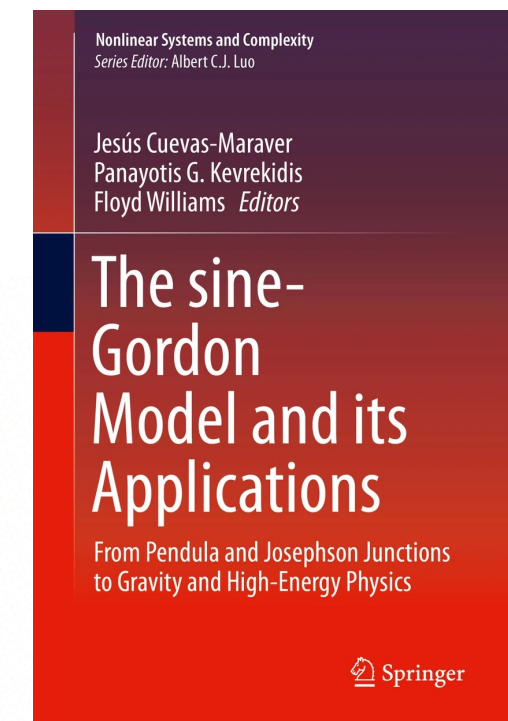
Controllable systems in laboratories:
cold atoms in lattices, ultracold gases,
ion trap, superconducting and photonic
circuits, ...

Simulating Quantum Field Theories (QFTs)

- * QFT is a ubiquitous framework for describing various phenomena in nature: particle physics, cosmology, critical systems,
- * Often fully “solving” QFT (i.e. obtain all correlation functions) is an extremely difficult task: **quantum field simulator** can be helpful
- * Prominent example: (Quantum) sine-Gordon model

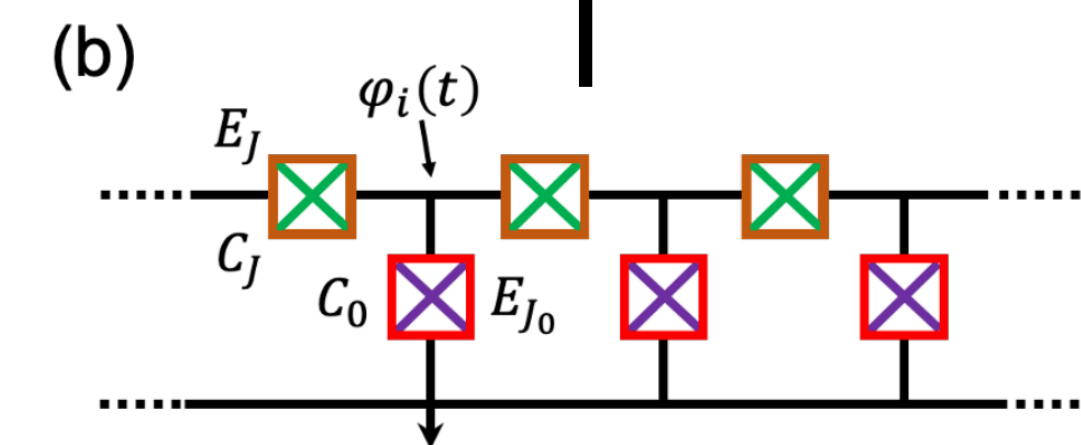
$$S_{\text{SG}}[\hat{\phi}_-] = \frac{\hbar K}{2\pi} \int d\mathcal{V} \sqrt{-g} \left[g^{\mu\nu} \partial_\mu \hat{\phi}_- \partial_\nu \hat{\phi}_- - \mathcal{M}^2 \cos \sqrt{2} \hat{\phi}_- \right].$$

Induces topological excitation,
i.e. solitons



Spin ladder

Wybo, E., Knap, M., & Bastianello, A..
PRB, 106(7), 075102 (2022).



Superconducting circuits

Roy, A., et al. Nuclear Physics B 968: 115445 (2021).

more
.....

Parallel 1D Ultracold Bose Gases

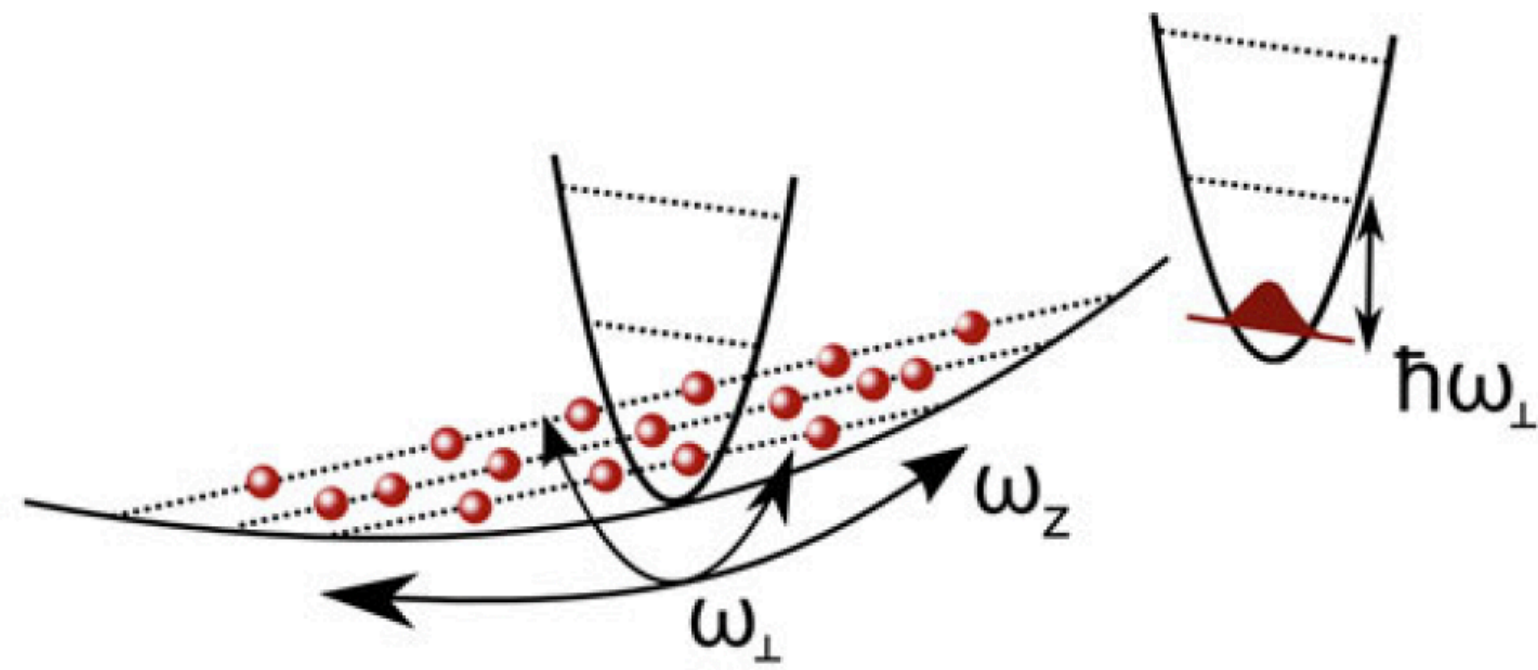
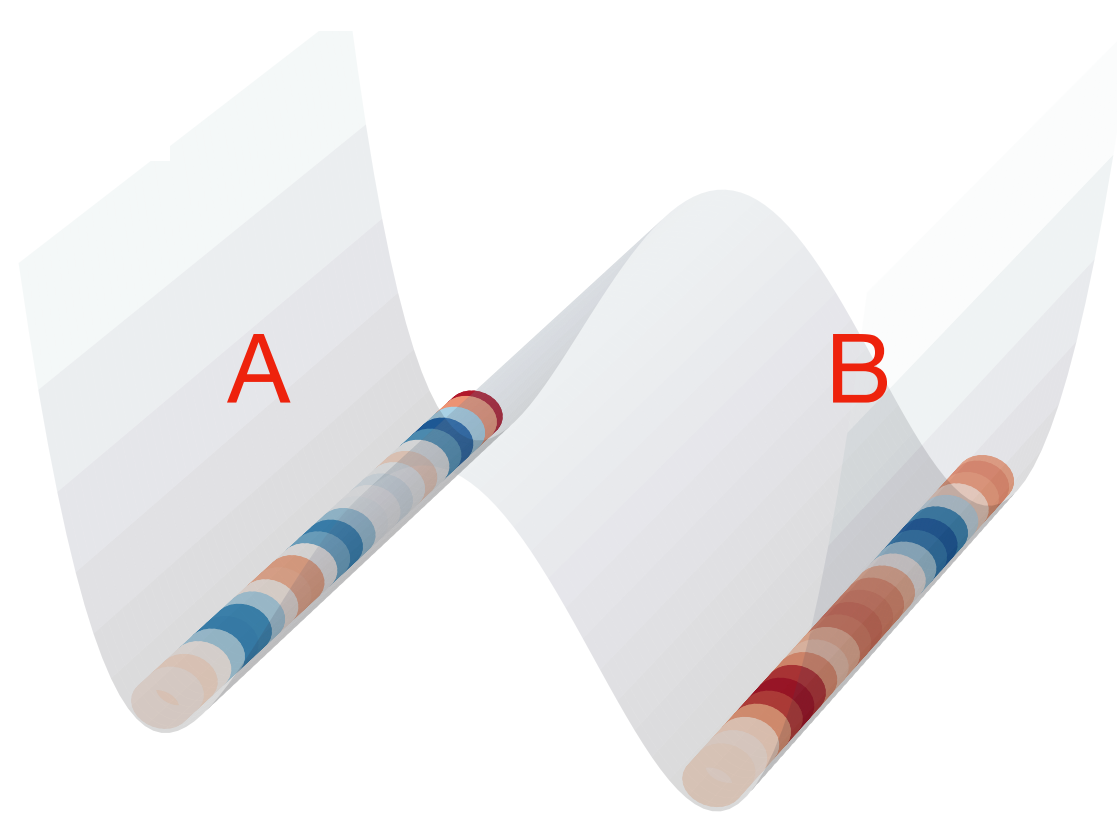


Image source: Tim Langen's PhD Thesis



$\hat{\Psi}_A(z), \hat{\Psi}_B(z)$: Bosonic field operators

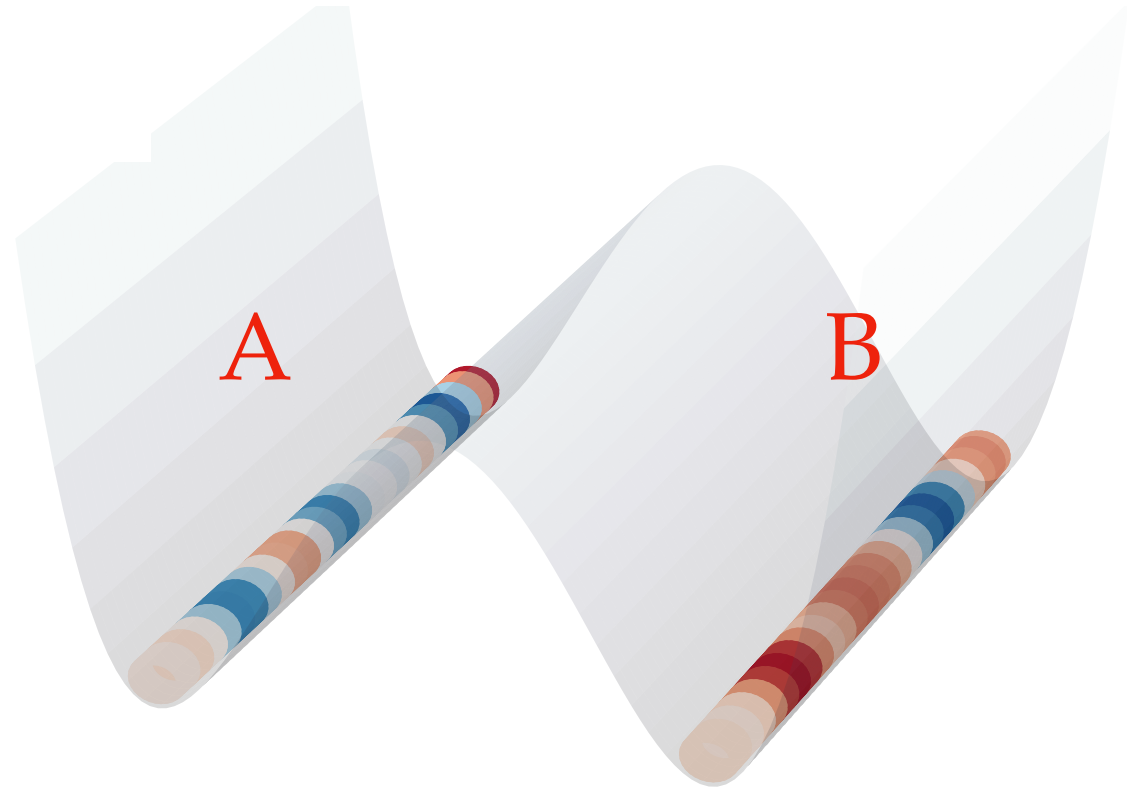
$$[\hat{\Psi}_\sigma(z), \hat{\Psi}_\kappa(z')] = \delta(z - z')\delta_{\sigma\kappa}$$

Microscopic model: tunnel-coupled Lieb-Liniger model

$$\hat{H} = \sum_{\sigma \in \{A, B\}} \left[\underbrace{-\frac{\hbar^2}{2m} \int \hat{\Psi}_\sigma^\dagger \partial_z^2 \hat{\Psi}_\sigma + (V(z) - \mu) \int \hat{\Psi}_\sigma^\dagger \hat{\Psi}_\sigma}_{\text{Kinetic + Potential}} + \underbrace{\frac{g}{2} \int \hat{\Psi}_\sigma^\dagger \hat{\Psi}_\sigma^\dagger \hat{\Psi}_\sigma \hat{\Psi}_\sigma}_{\text{Contact Interaction}} \right] - \hbar J \int dz \left[\hat{\Psi}_A^\dagger \hat{\Psi}_B + \text{h.c.} \right]_{\text{Tunnel-coupling}}$$

Gritsev, V., Polkovnikov, A., & Demler, E. Linear response theory for a pair of coupled one-dimensional condensates of interacting atoms. PRBs, 75(17), 174511 (2007).

Effective Low-Energy Model



$$\hat{\Psi}_\sigma(z) = e^{i\hat{\phi}_\sigma(z)} \sqrt{n_0(z) + \delta\hat{n}_\sigma(z)}$$

$$[\delta\hat{n}_\sigma(z), \hat{\phi}_\kappa(z')] = i\delta(z - z')\delta_{\sigma\kappa}$$

Relative (-) and common (+) fields

$$\delta\hat{n}_\pm(z) = \delta\hat{n}_A(z) \pm \delta\hat{n}_B(z)$$

$$\hat{\phi}_\pm(z) = \hat{\phi}_A(z) \pm \hat{\phi}_B(z)$$

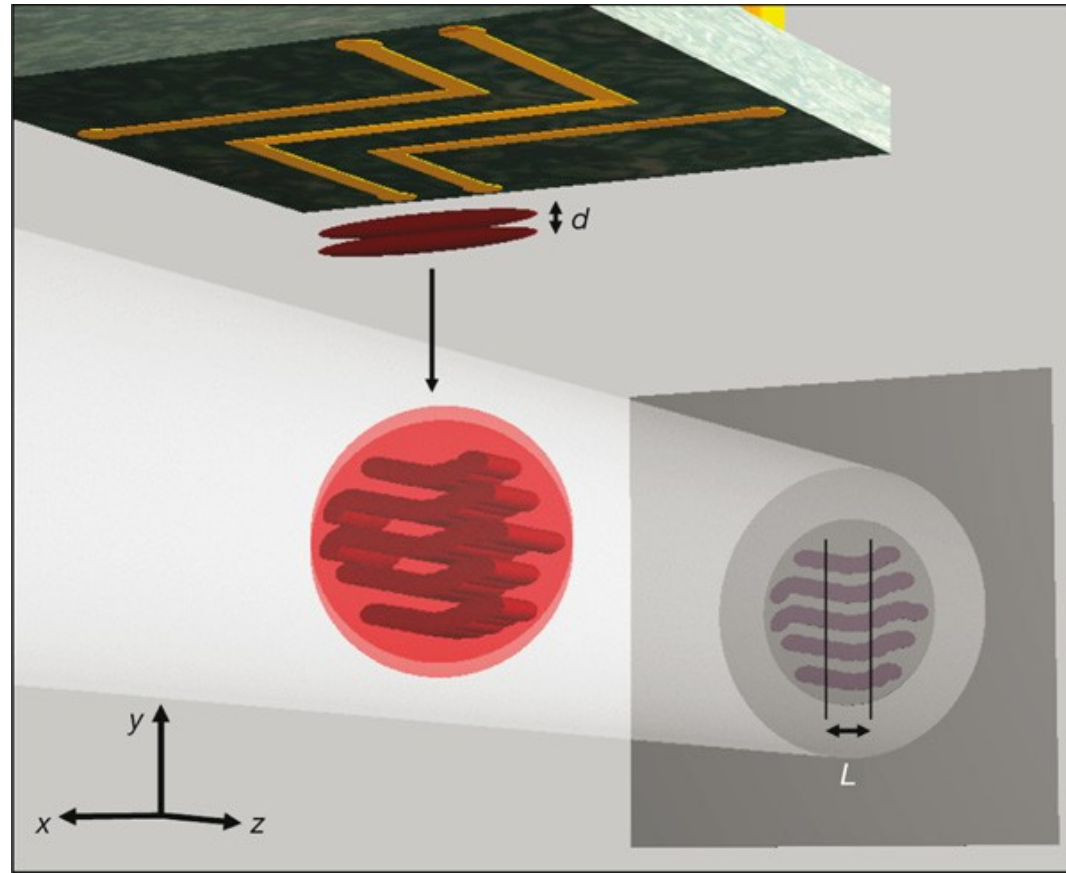
$$\hat{H} \approx \hat{H}_{SG} = \hat{H}_{TLL}[\delta\hat{n}_+, \hat{\phi}_+] + \hat{H}_{TLL}[\delta\hat{n}_-, \hat{\phi}_-] - 2\hbar J \int n_0(z) \cos \hat{\phi}_-(z) dz$$

$$\hat{H}_{TLL}[\delta\hat{n}, \hat{\phi}] = \frac{\hbar c}{2} \int \left[\frac{\pi}{K} (\delta\hat{n}(z))^2 + \frac{K}{\pi} (\partial_z \hat{\phi}(z))^2 \right] dz$$

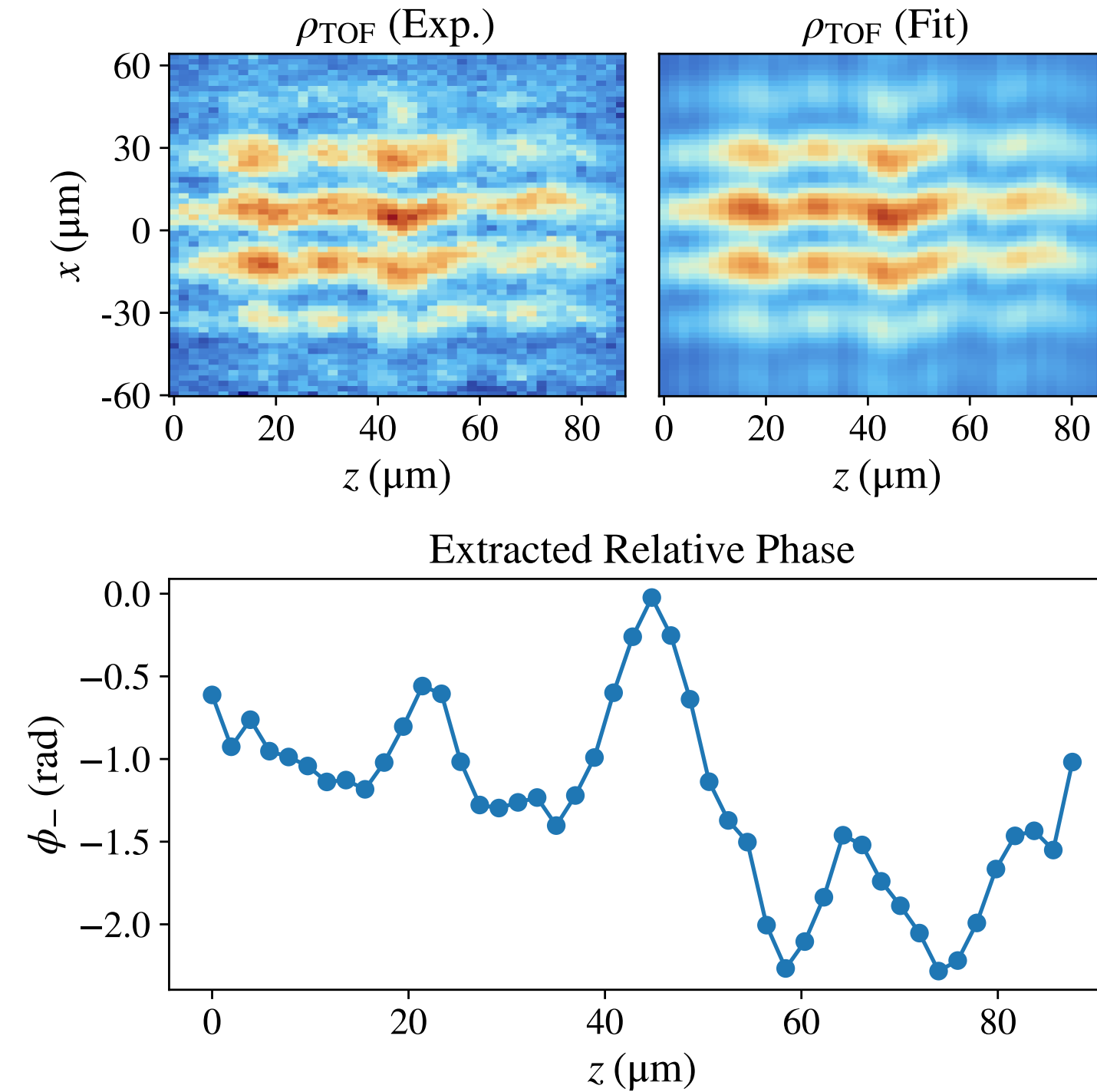
sine-Gordon
interaction

Tomonaga-Luttinger Liquid (TLL)

Matter-wave Interference Readout



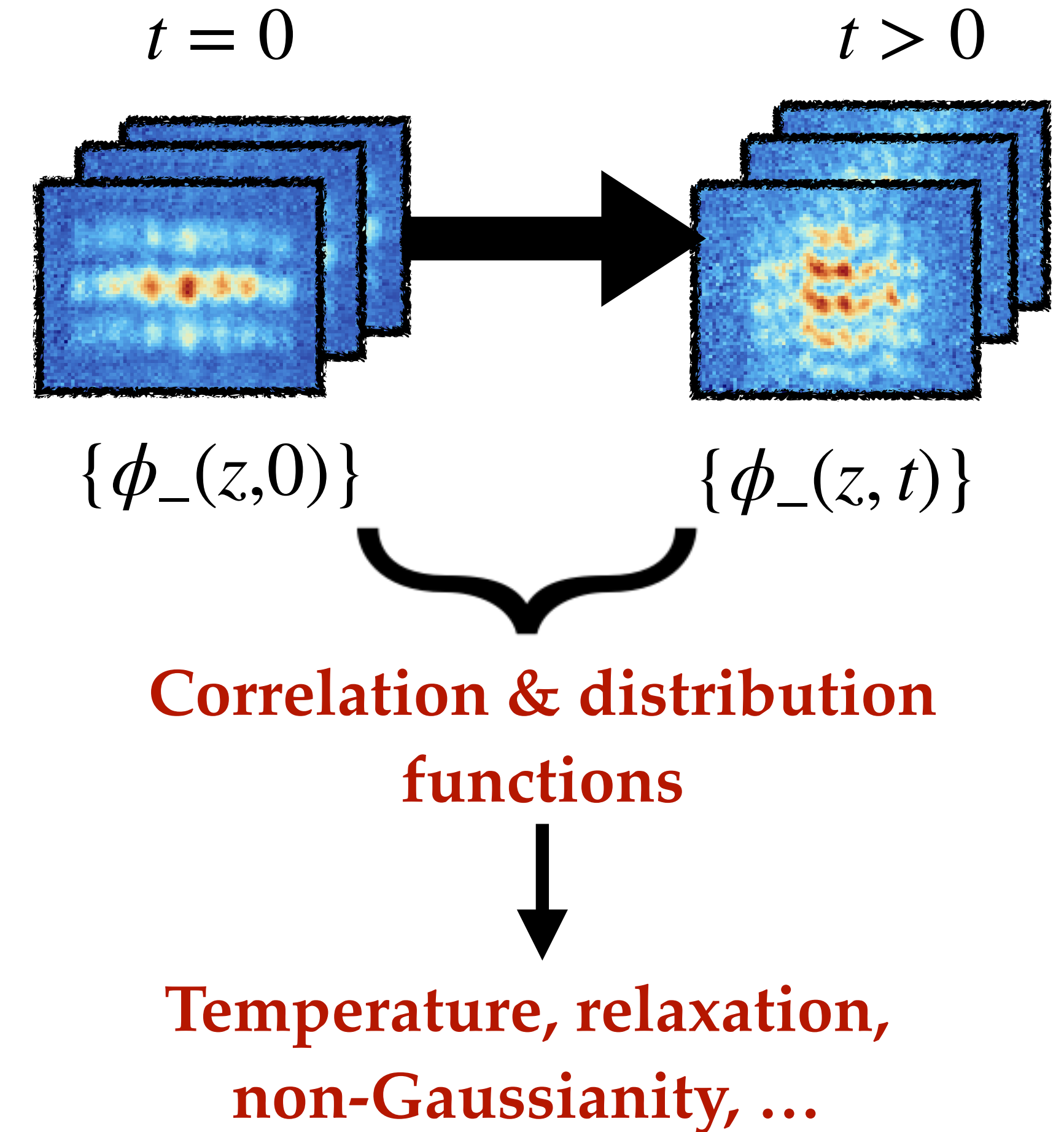
Hofferberth, S., Lesanovsky, I., Fischer, B., Schumm, T., & Schmiedmayer, J.
Nature, 449(7160), 324-327. (2007)



$$\rho_{\text{TOF}}(x, z, \tau) = A(z)e^{-x^2/\sigma_\tau^2} \left[1 + B(z) \cos(kx + \phi_-(z)) \right]$$

van Nieuwkerk, Y. D, Schmiedmayer, J. and Essler, F. *SciPost Physics* 5.5: 046 (2018).

Murtadho, T., et al. *SciPost Physics* 18.2: 065 (2025).



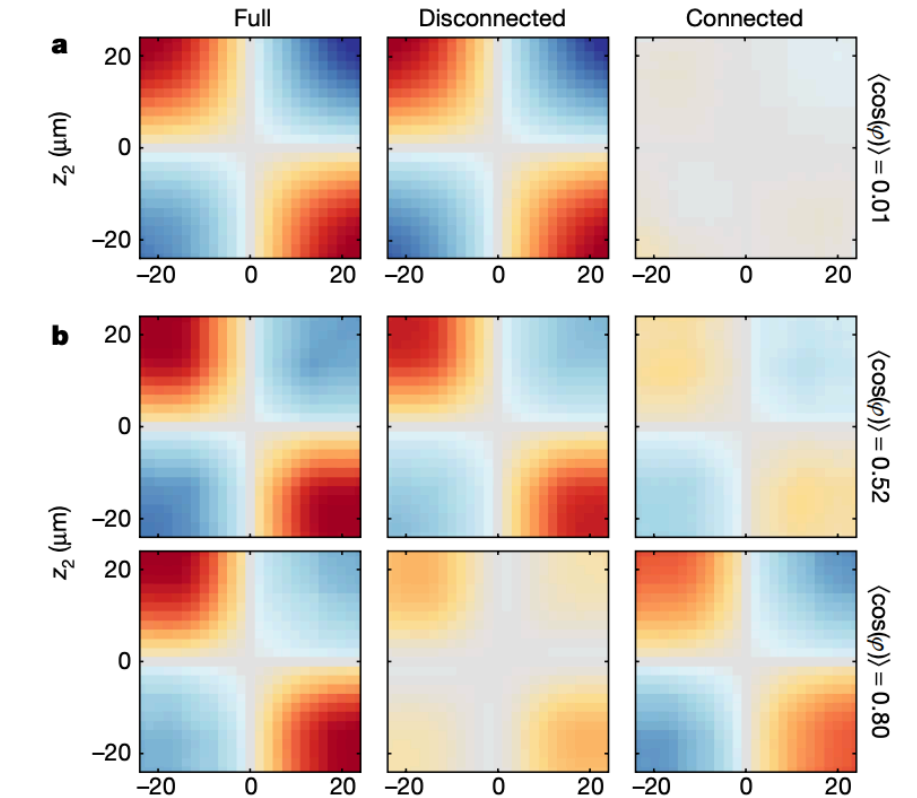
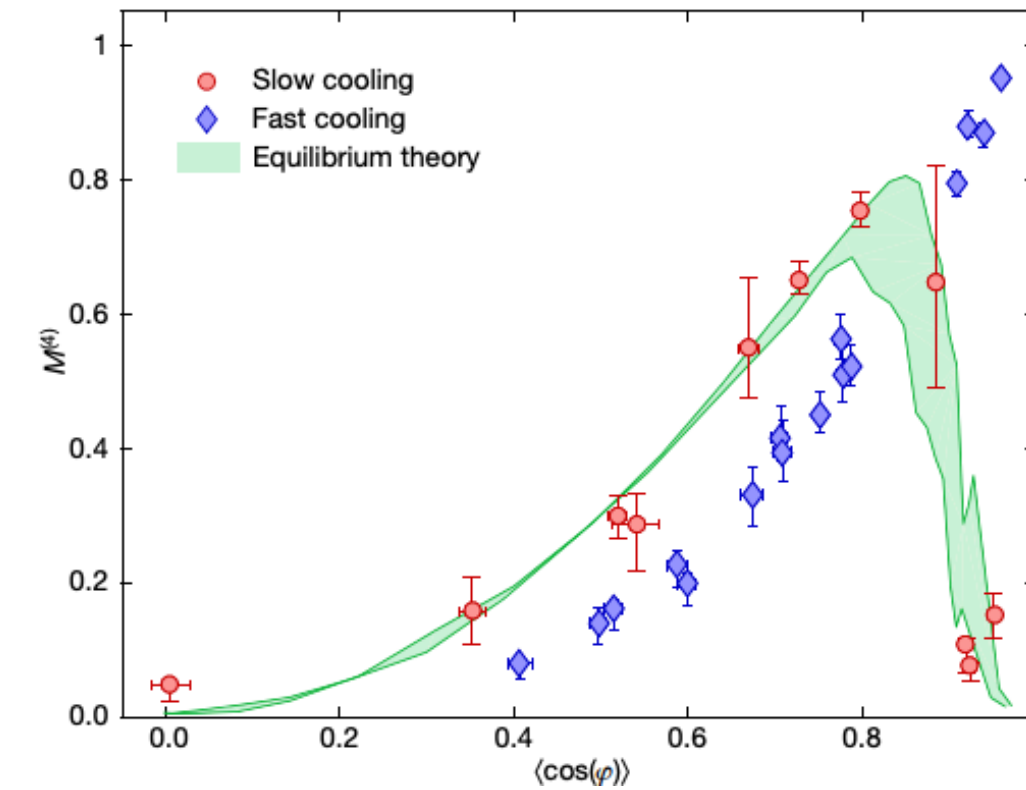
Simulating sine-Gordon model

Investigate **non-Gaussian** statistics

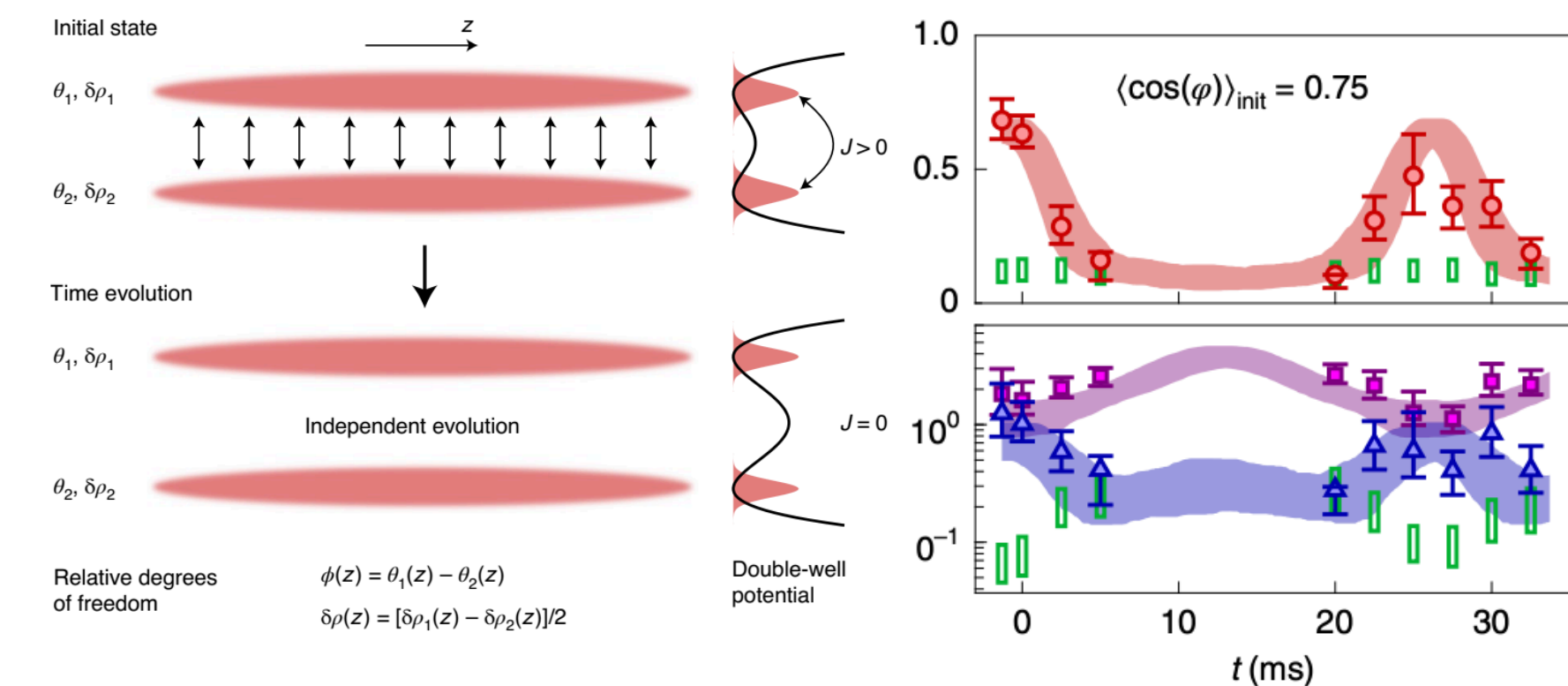
$$G_N(\mathbf{z}; z_0) = \left\langle \prod_{i=1}^N (\hat{\phi}_-(z_i) - \hat{\phi}_-(z_0)) \right\rangle$$

$$G_N(\mathbf{z}; z_0) = G_N^{(\text{disc.})}(\mathbf{z}; z_0) + G_N^{(\text{con.})}(\mathbf{z}; z_0)$$

$$M^{(N)} = \frac{\sum_{\mathbf{z}} |G_N^{(\text{con.})}(\mathbf{z}; z_0)|}{\sum_{\mathbf{z}} |G_N(\mathbf{z}; z_0)|} \quad N^{\text{th}} \text{ order non-Gaussian measure}$$



Schweigler, T., et al. "Experimental characterization of a quantum many-body system via higher-order correlations." *Nature* 545.7654: 323-326 (2017).



Schweigler, T., et al. "Decay and recurrence of non-Gaussian correlations in a quantum many-body system." *Nat. Phys.* 17.5: 559-563 (2021).

Gaussian Tomography

$$[\delta\hat{n}_-(z), \hat{\phi}_-(z')] = i\delta(z - z')$$

Gaussian state $\leftrightarrow$ **Covariance matrix** Γ

$$\delta\hat{n}_-(z) = \sum_{\mu} f_{\mu}(z) \delta\hat{n}_{-}^{\mu} \quad \hat{\phi}_-(z) = \sum_{\mu} f_{\mu}(z) \hat{\phi}_{-}^{\mu}$$

$$\Gamma^{-} = \begin{pmatrix} \Gamma_{\delta n \delta n}^{-} & \Gamma_{\delta n \phi}^{-} \\ \Gamma_{\phi \delta n}^{-} & \Gamma_{\phi \phi}^{-} \end{pmatrix}$$

$$\hat{H}_{\text{TLL}}^{-} = \frac{\hbar c}{2} \int \left[\frac{\pi}{K} (\delta\hat{n}_-(z))^2 + \frac{K}{\pi} (\partial_z \hat{\phi}_-(z))^2 \right] dz$$

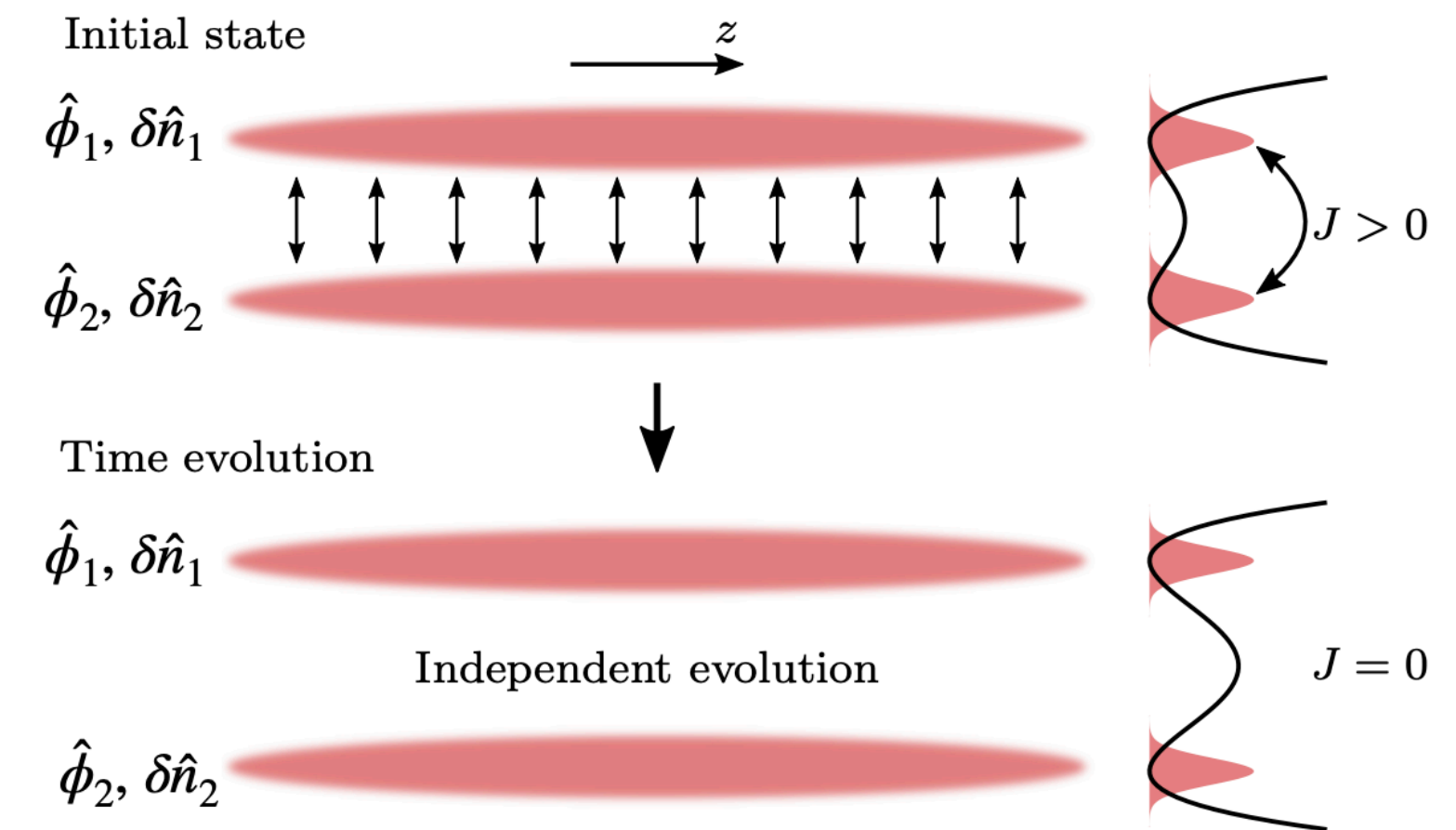
TOF Measurement

$$\{ \{ \phi_-(z, t_1) \}, \{ \phi_-(z, t_2) \}, \dots \}$$

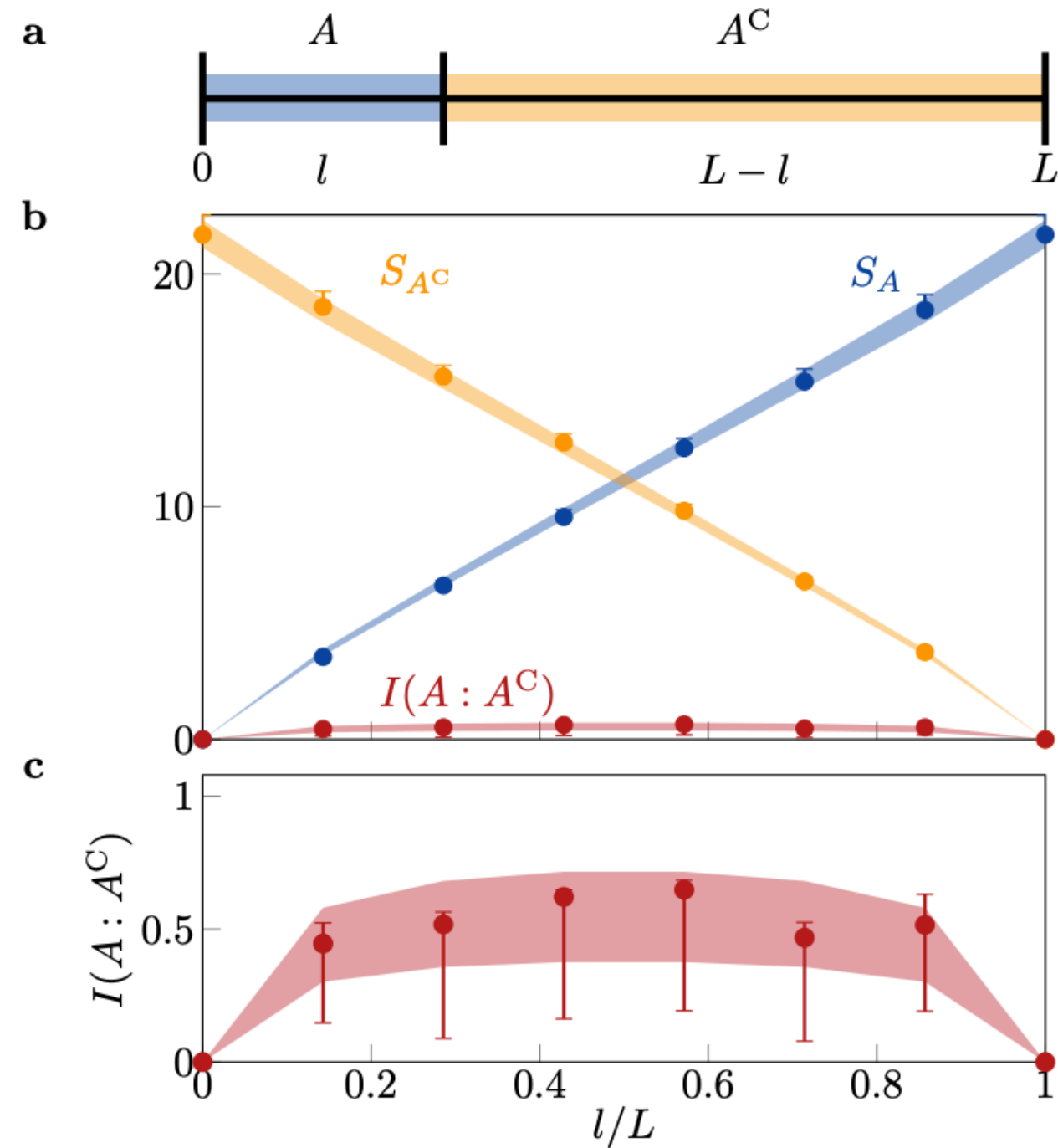
Tomography

Gluza, M., et al. "Quantum read-out for cold atomic quantum simulators." Communications Physics 3.1: 12 (2020).

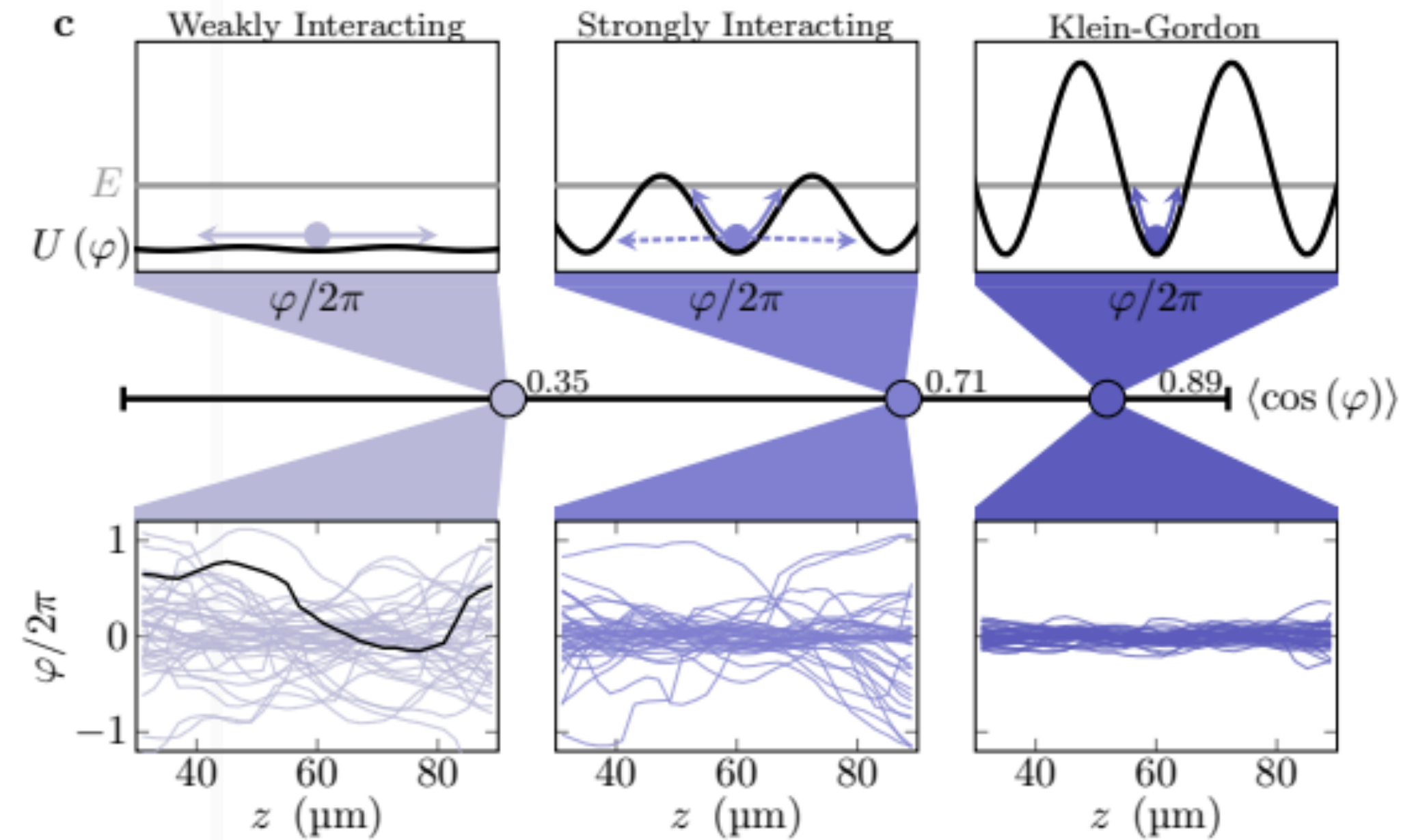
Information-theoretic quantities: **entropies, mutual information,**



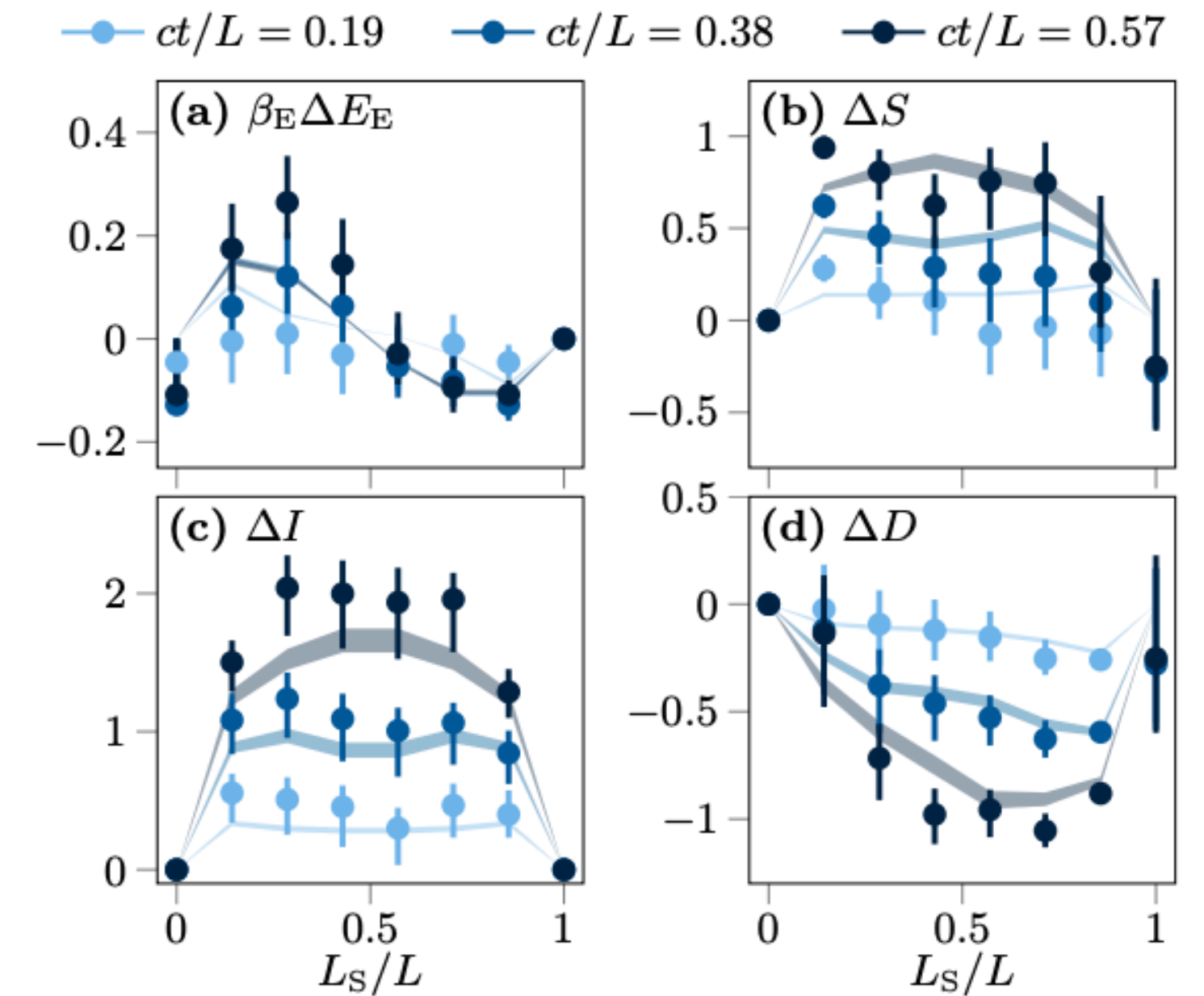
Quantum field theory x quantum information



Tajik, Mohammadamin, et al. "Verification of the area law of mutual information in a quantum field simulator." *Nature Physics* 19.7: 1022-1026 (2023).



Jarema, Maciej T., et al. "Observation of area laws in an interacting quantum field simulator." *arXiv:2510.13783* (2025).



Aimet, Stefan, et al. "Experimentally probing Landauer's principle in the quantum many-body regime." *Nature Physics* 21.8: 1326-1331 (2025).

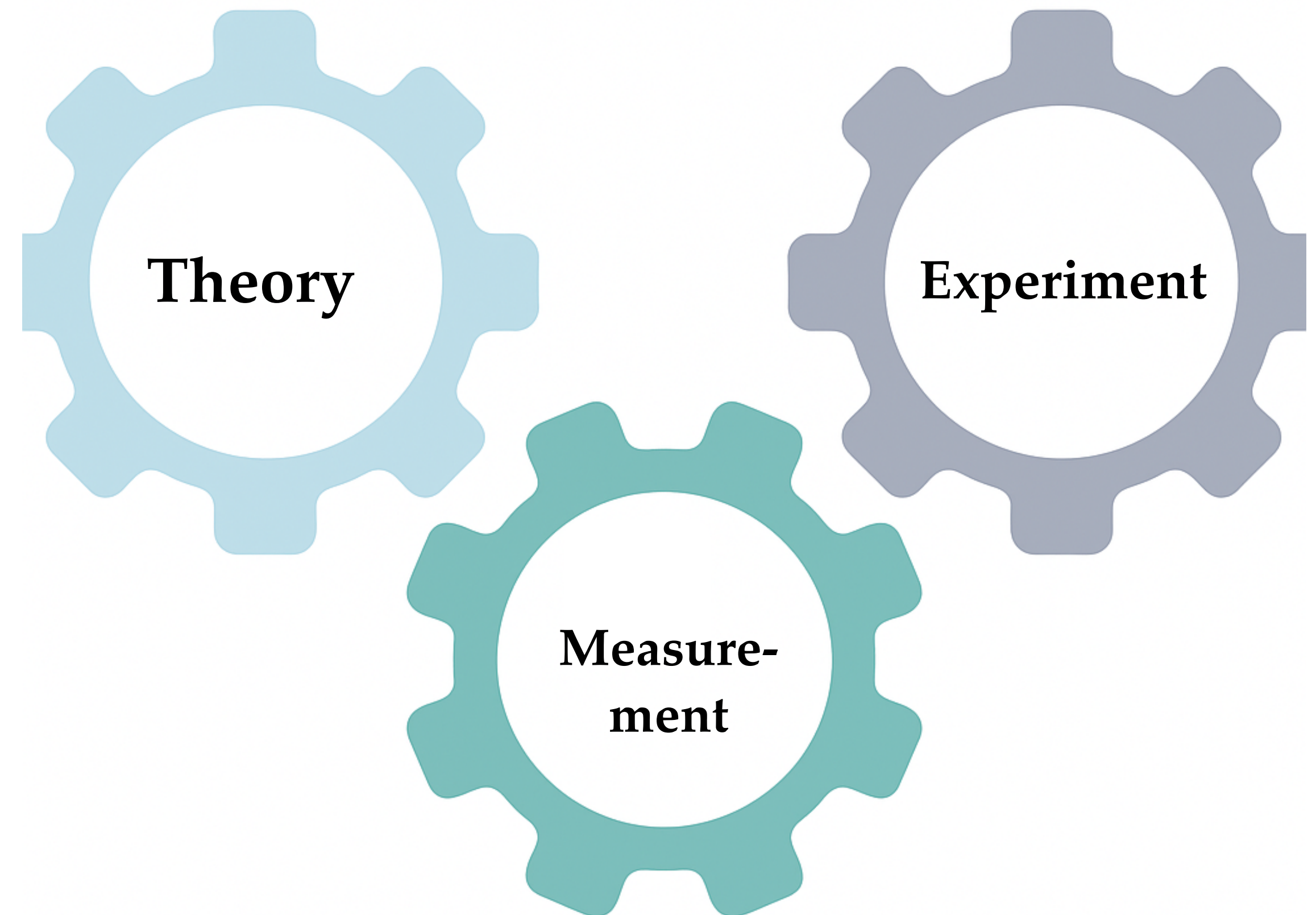
Entanglement??



Keep alive
quantum mechanics!

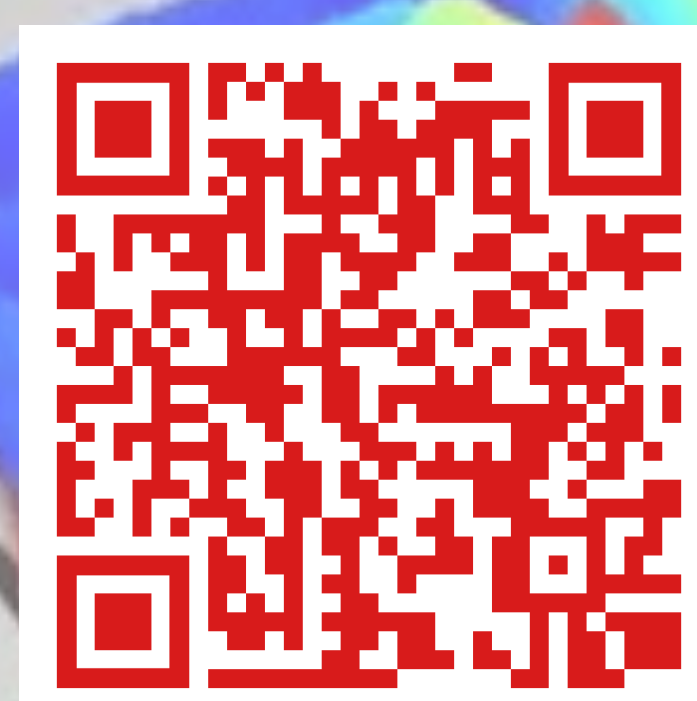
Measurement bottleneck

- * **Limited available measurements** create a bottleneck to the physical phenomena that can be probed in parallel 1D Bose gases
- * **Research aim: expand the measurement toolbox** and probe **new physical phenomena** in quantum field simulators (e.g. entanglement)



Results

Common Phase Extraction → Full Tomography
→ Field Entanglement at Finite Temperatures



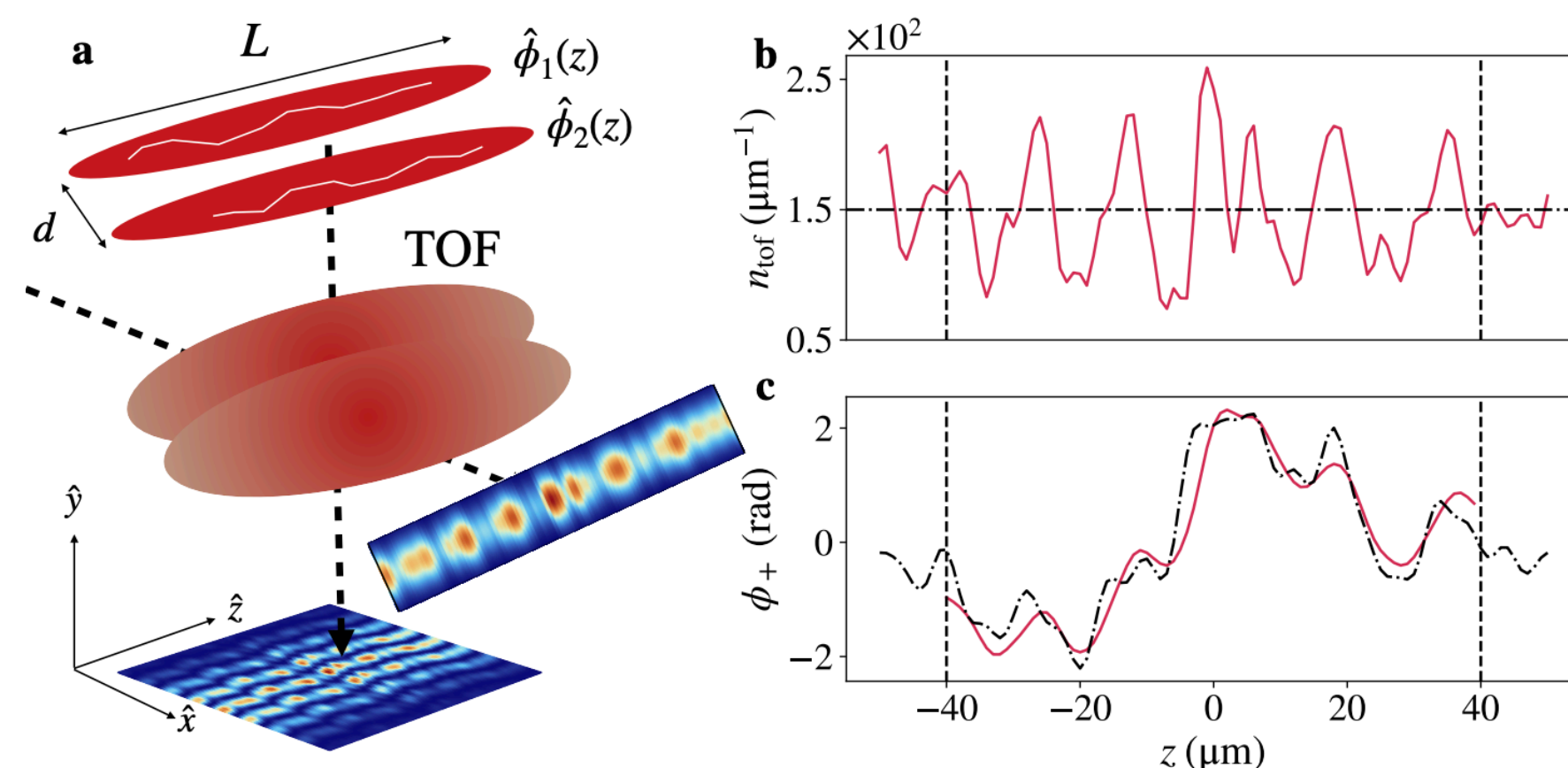
Measurement of common phase

$$\hat{\phi}_+(z) = \hat{\phi}_1(z) + \hat{\phi}_2(z)$$

Continuity equation during TOF dynamics

$$\partial_t \hat{n}_+ + \frac{\hbar}{2m} \nabla \cdot (n_0 \nabla \hat{\phi}_+) = 0$$

$$\partial_z \hat{\phi}_+(z - dz) \longrightarrow \partial_z \hat{\phi}_+(z + dz)$$



PHYSICAL REVIEW RESEARCH 7, L022031 (2025)

Letter



Measurement of total phase fluctuation in cold-atomic quantum simulators

Taufiq Murtadho^{1,*}, Federica Cataldini², Sebastian Erne², Marek Gluza¹, Mohammadamin Tajik², Jörg Schmiedmayer², and Nelly H. Y. Ng^{1,†}

¹School of Physical and Mathematical Sciences, *Nanyang Technological University*, Singapore 639673

²Vienna Center for Quantum Science and Technology, Atominstitut, TU Wien, Stadionallee 2, 1020 Vienna, Austria

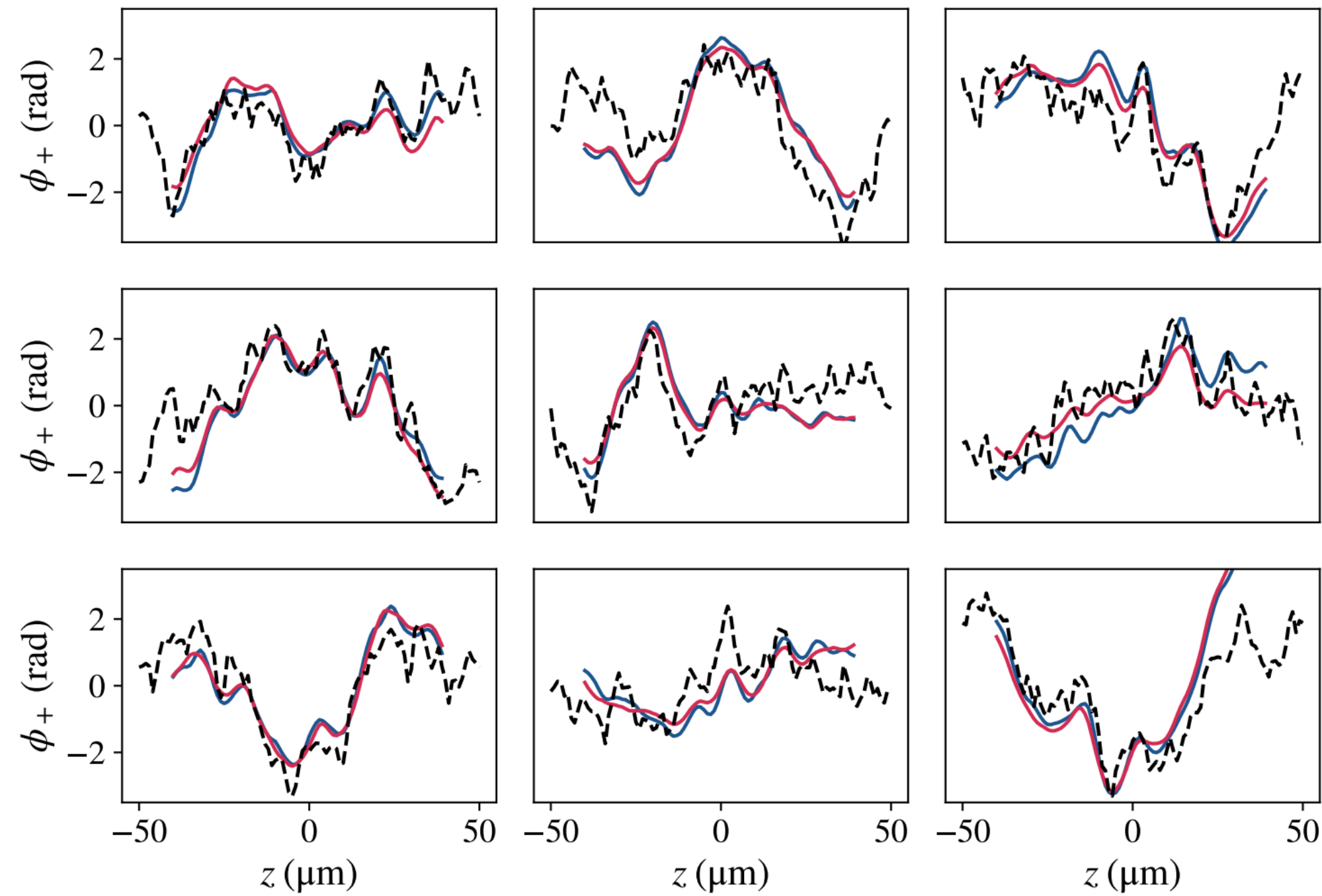


(Received 18 October 2024; accepted 28 February 2025; published 7 May 2025)

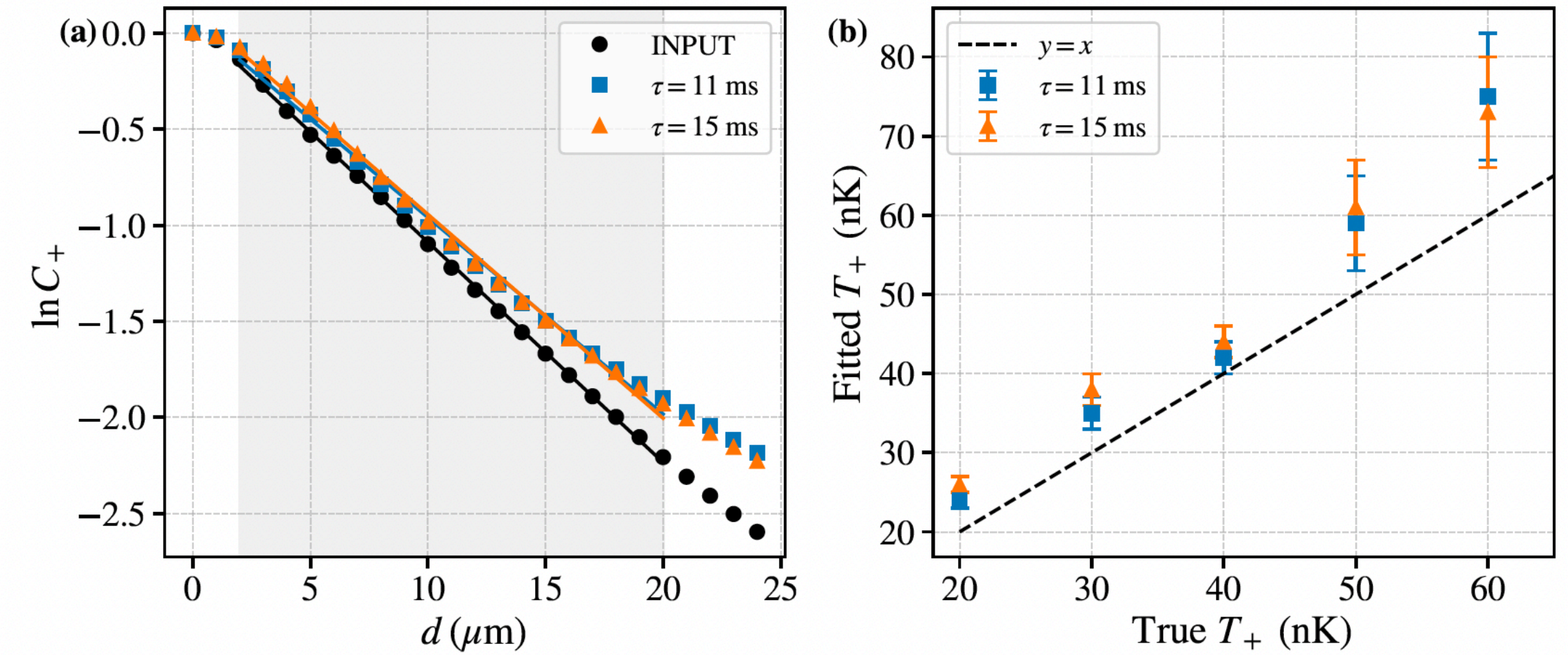
Common phase recon. $\approx$ Solving **Poisson's equation**

$$\partial_z^2 \hat{\phi}_+ = \frac{1}{\ell_\tau^2} \left(1 - \frac{n_{\text{tof}}(z, \tau)}{n_0(z)} \right)$$

Numerical Benchmarking (1)



Single shot examples



**Vertex correlation
function**

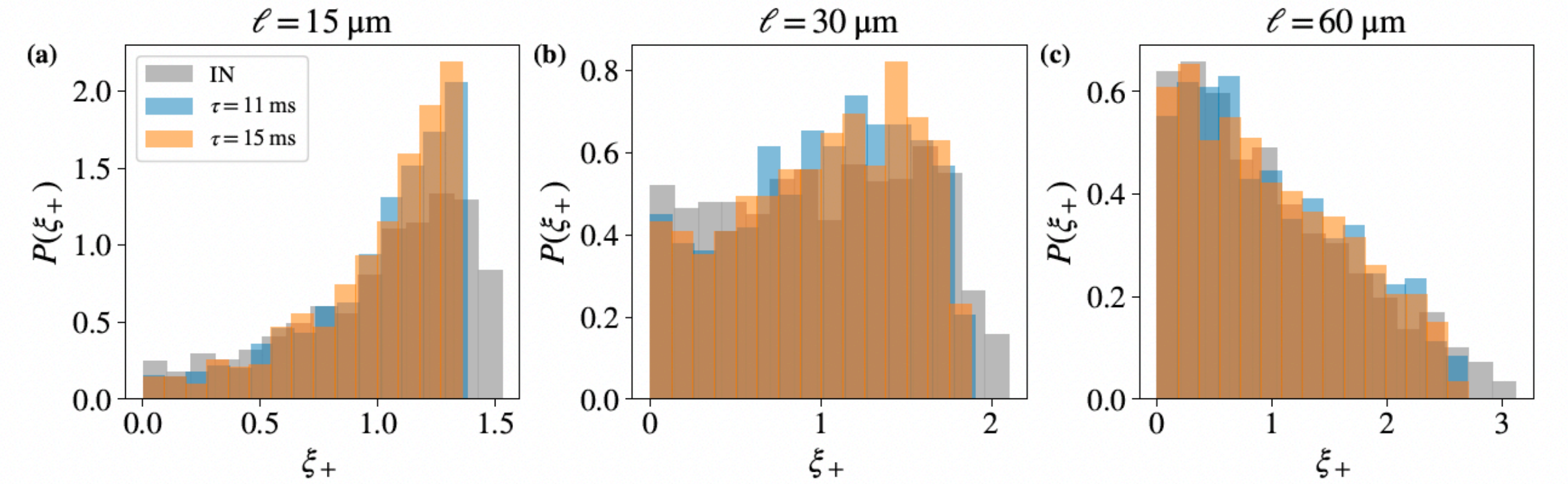
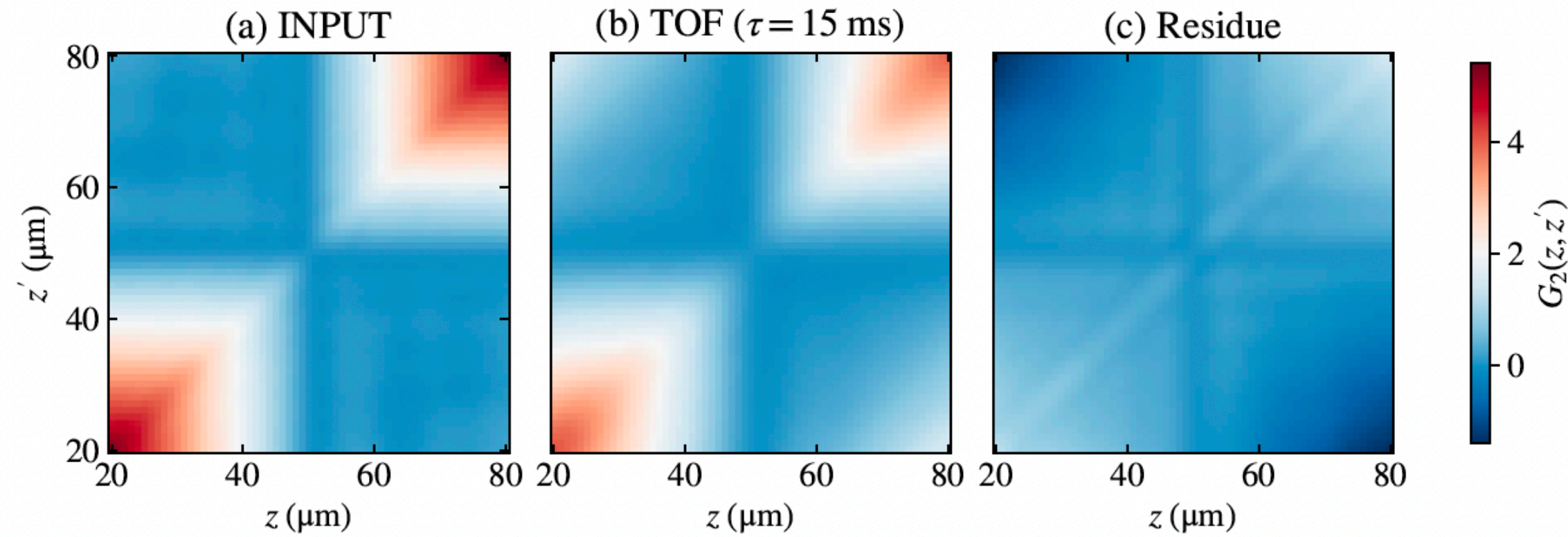
Thermometry

$$C_+(z, z') = \left\langle \exp \left[i(\phi_+(z) - \phi_+(z')) \right] \right\rangle$$

$$\sim \exp(-|z - z'|/\lambda_{T_+})$$

$$\lambda_{T_+} = \frac{\hbar^2 n_0}{m k_B T_+}$$

Numerical Benchmarking (2)



Two-point function

$$G_+^{(2)}(z, z') = \langle [\phi_+(z) - \phi_+(0)][\phi_+(z') - \phi_+(0)] \rangle$$

Can be generalized to higher-order, i.e. $G_+^{(N)}(\mathbf{z})$

Contrast Distribution Function

$$\xi_+(\ell) = \frac{\left| \int_{-\ell/2}^{\ell/2} \exp(i\phi_+(z)) dz \right|^2}{\left\langle \left| \int_{-\ell/2}^{\ell/2} \exp(i\phi_+(z)) dz \right|^2 \right\rangle}$$

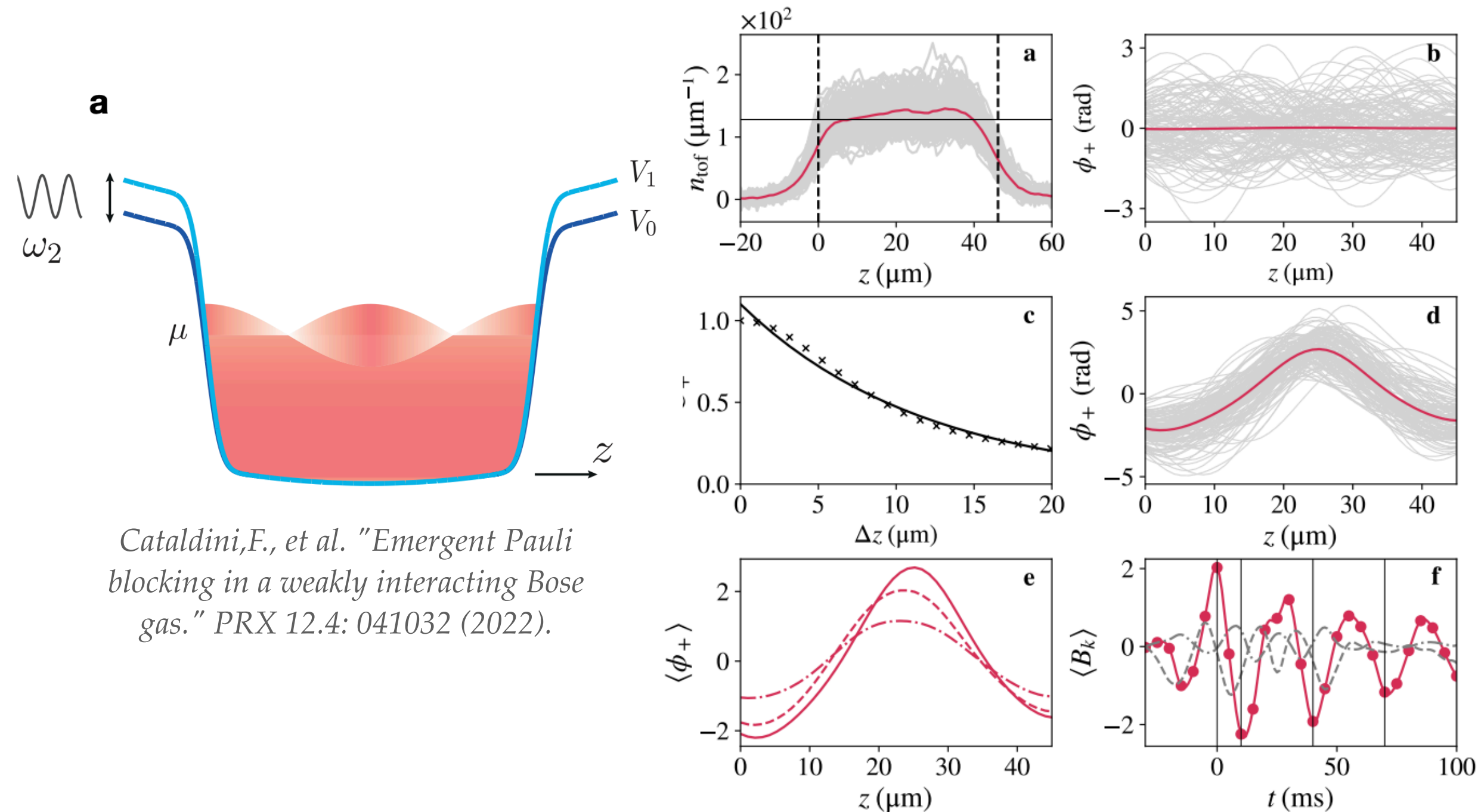
Gluza, M., et al. "Quantum read-out for cold atomic quantum simulators." *Communications Physics* 3.1: 12 (2020).

Adu Smith, D., et al. "Prethermalization revealed by the relaxation dynamics of full distribution functions." *NJP* 15.7: 075011 (2013).

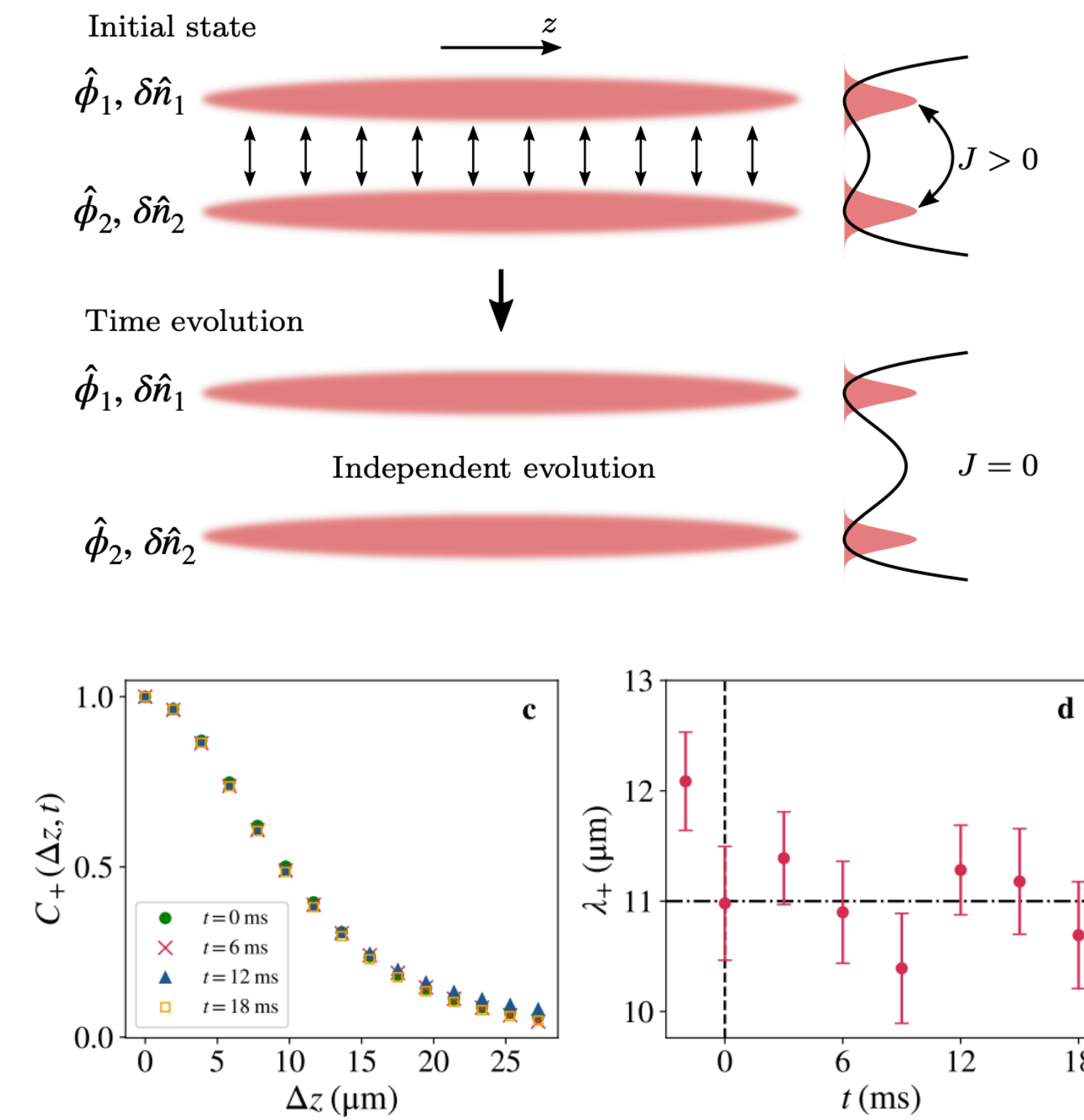
All correlation functions probed for relative phase are now accessible for the common phase!

Experimental Benchmarking

Exp. 1: Relaxation of Driven Luttinger Liquids

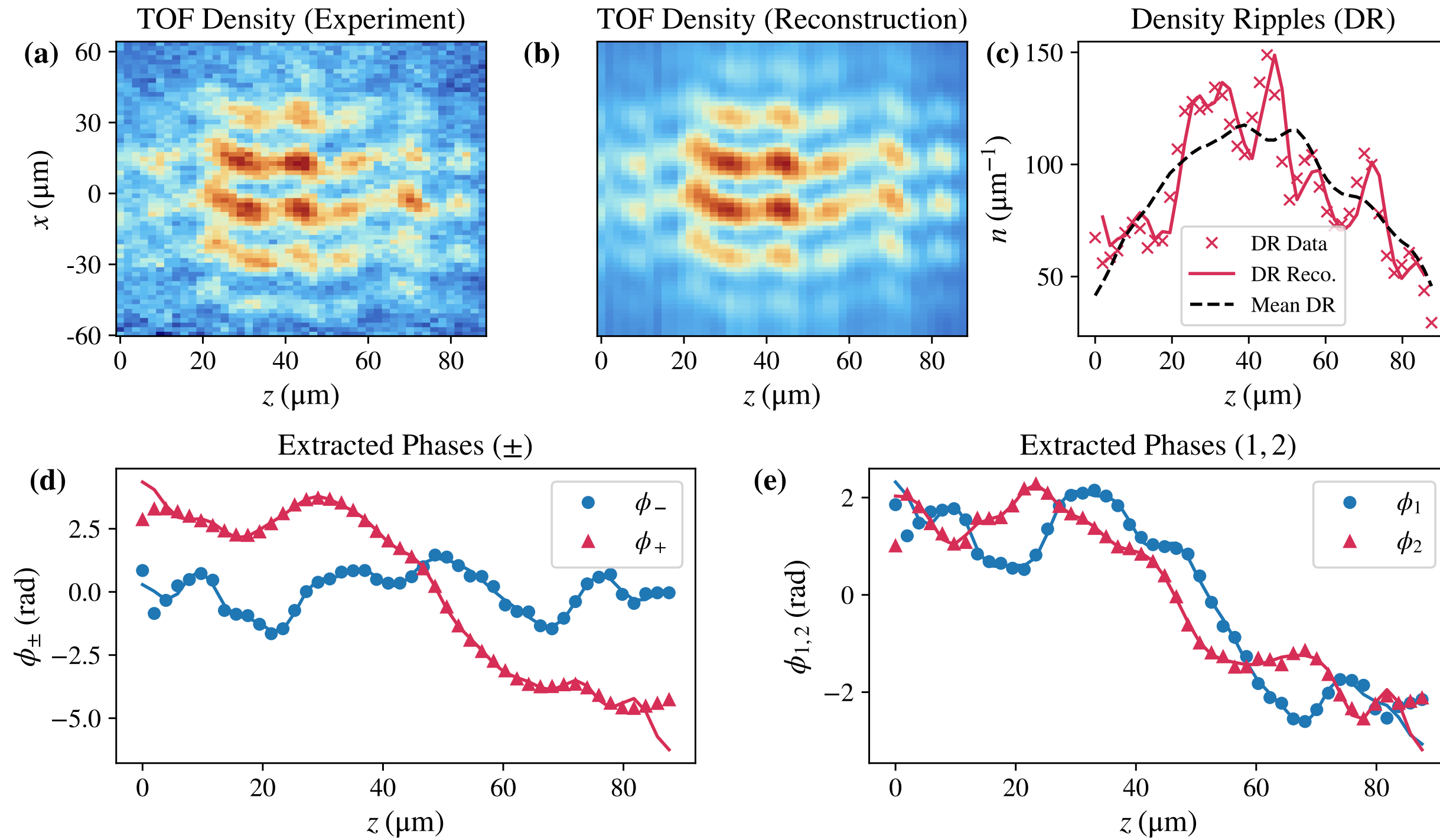


Exp. 2: Dynamics after tunneling quench



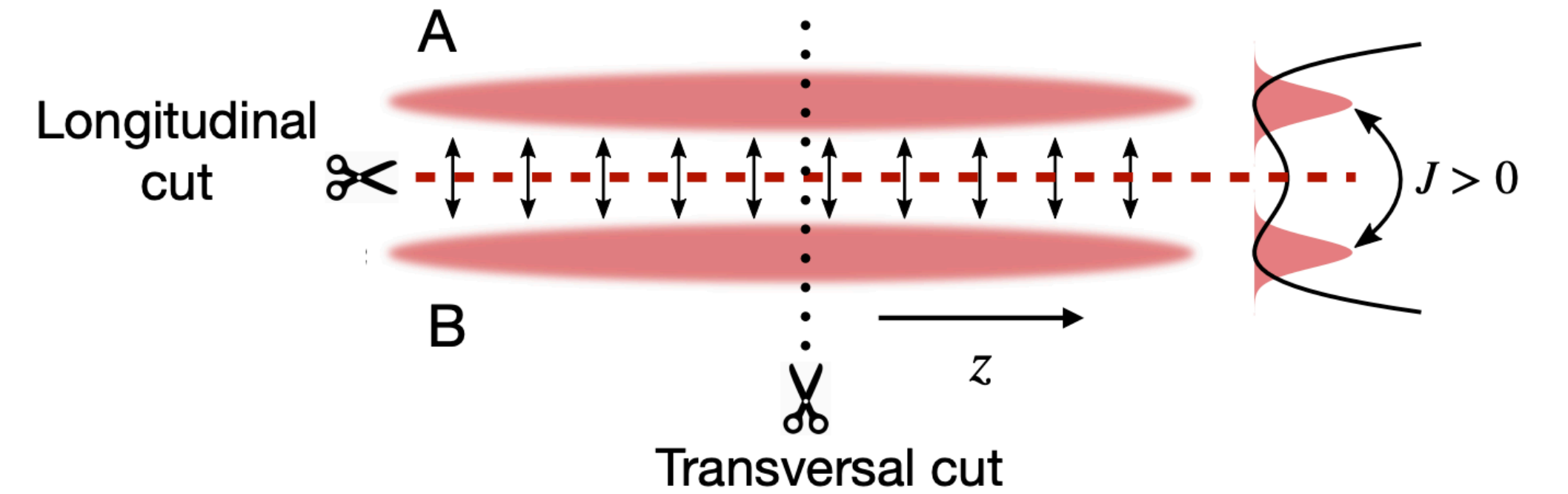
Probe $C_+(z, t)$
after quench
from $J > 0$ to
 $J = 0$

Expanding the Space of Observables



$$\phi_{1,2}(z) = \frac{1}{2}[\phi_+(z) \pm \phi_-(z)]$$

(a) Tunnel coupling

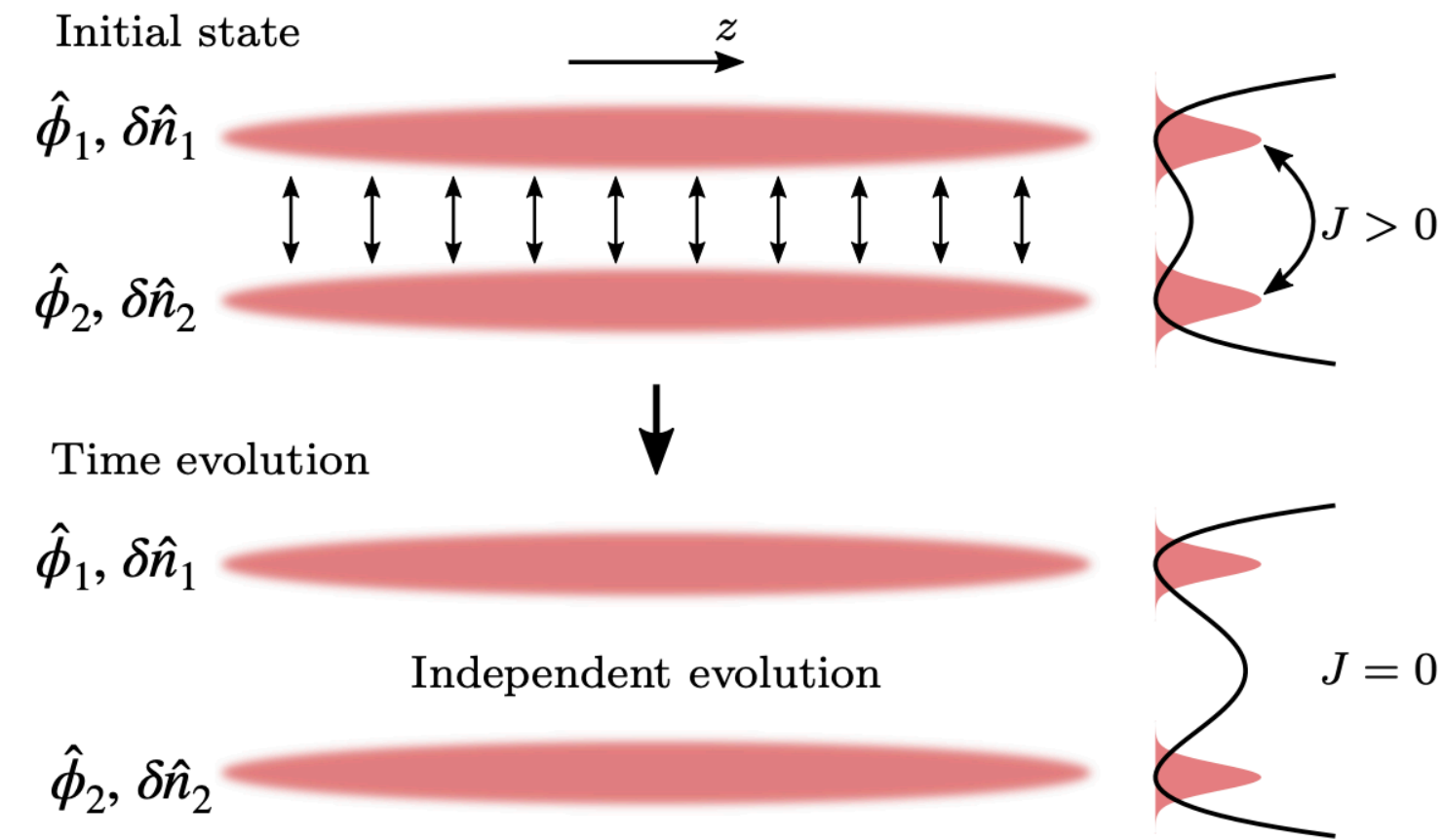


Direct access toward left-right correlation, e.g. $\langle \phi_1(z) \phi_2(z') \rangle$

Full Tomography

$$\Gamma = \begin{pmatrix} \Gamma_+ & 0 \\ 0 & \Gamma_- \end{pmatrix} \rightarrow \begin{pmatrix} \Gamma_{AA} & \Gamma_{AB} \\ \Gamma_{BA} & \Gamma_{BB} \end{pmatrix}$$

Full Gaussian Tomography



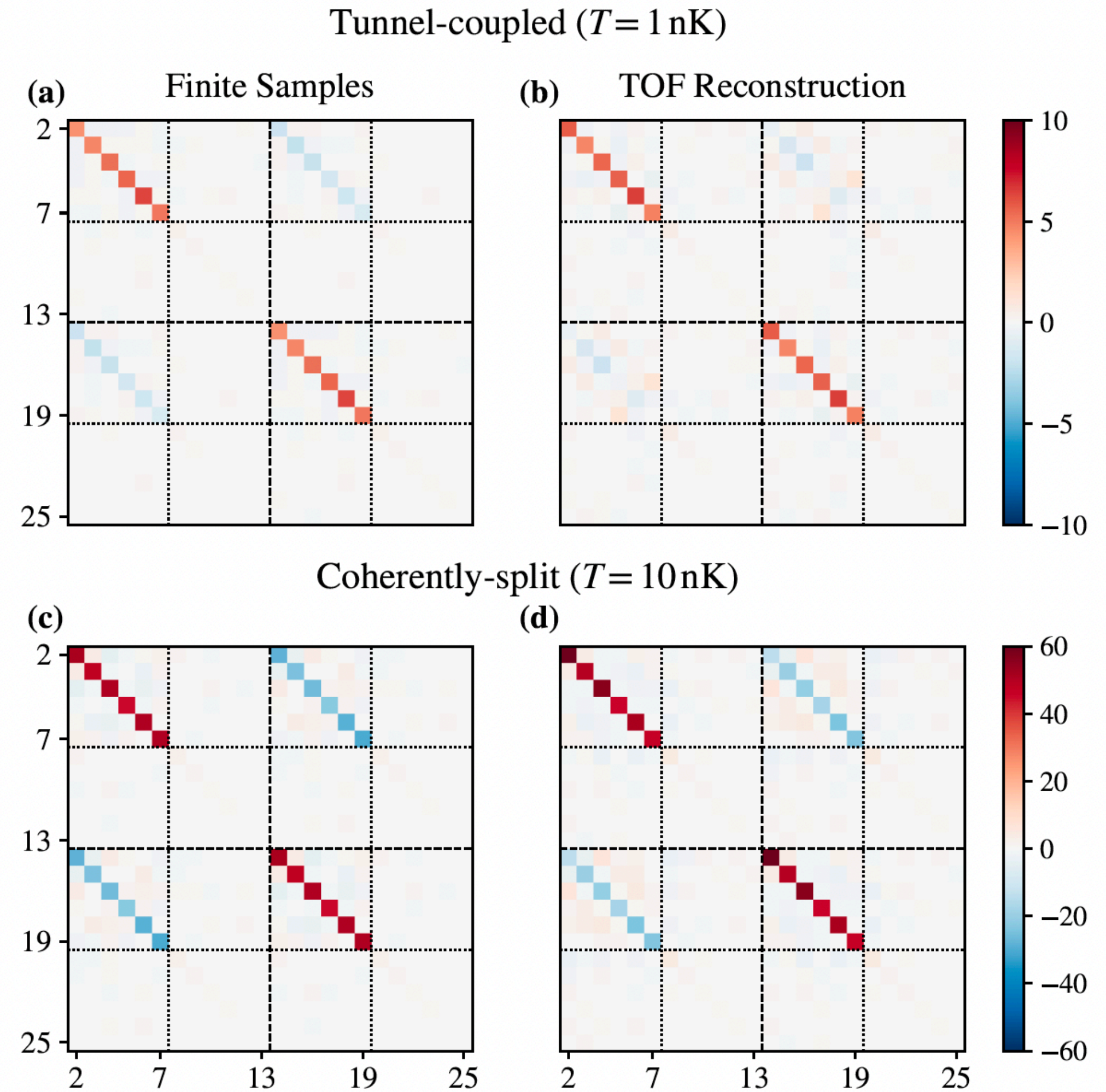
$$\hat{H}_{\text{TLL}}^{\pm} = \frac{\hbar c}{2} \int \left[\frac{\pi}{K} (\delta\hat{n}_{\pm}(z))^2 + \frac{K}{\pi} (\partial_z \hat{\phi}_{\pm}(z))^2 \right] dz$$

↓ TOF Measurement

$$\{ \{ \phi_{\pm}(z, t_1) \}, \{ \phi_{\pm}(z, t_2) \}, \dots \}$$

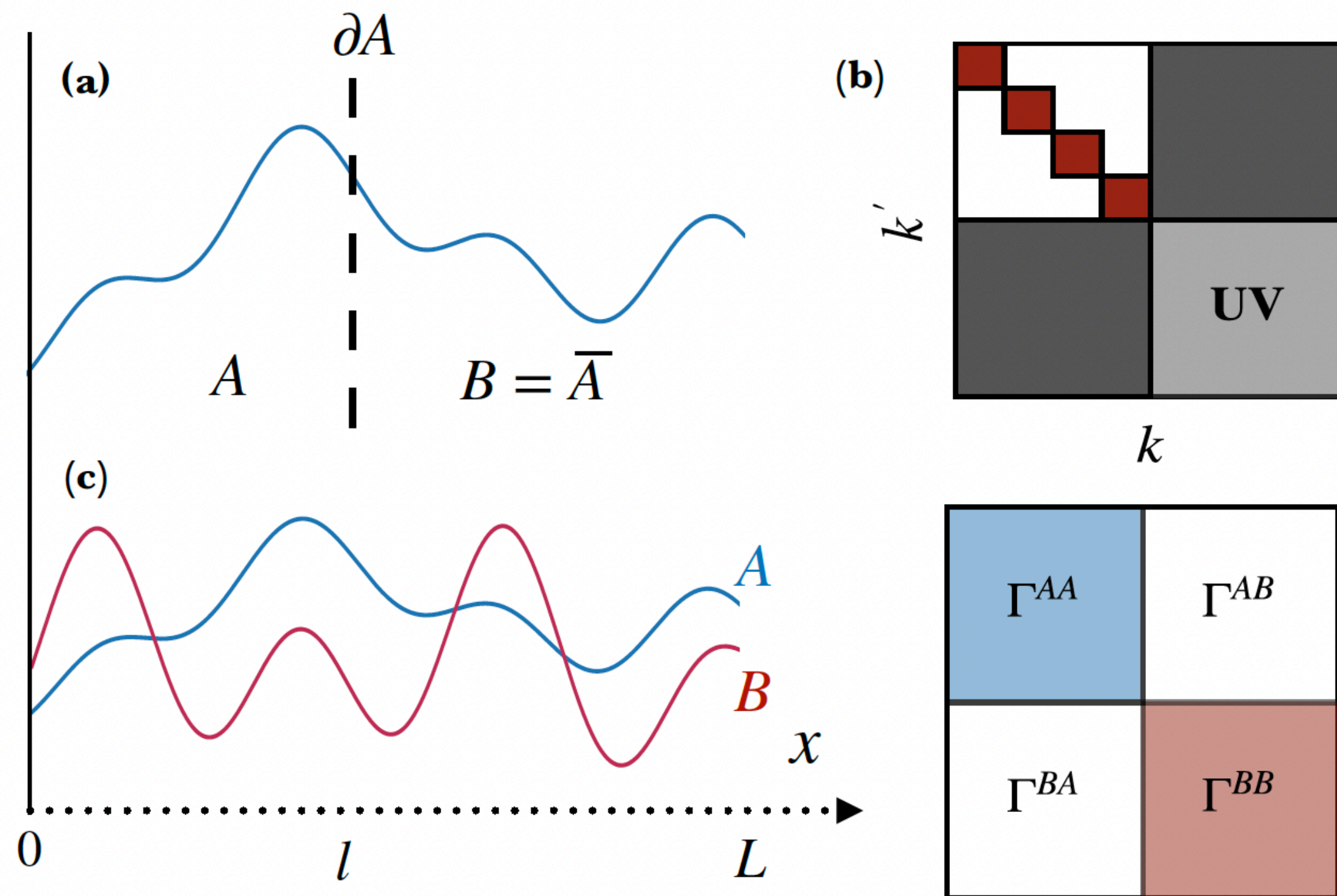
↓ Full Tomography

$$\Gamma = \begin{pmatrix} \Gamma_{AA} & \Gamma_{AB} \\ \Gamma_{BA} & \Gamma_{BB} \end{pmatrix}$$



Entanglement in Quantum Fields

Entanglement depends on the **partition**



* Real-space entanglement

Area law of entanglement

Calabrese, P., and Cardy, J. "Entanglement entropy and quantum field theory." JSTAT 2004.06: P06002 (2004).

.....

* Momentum-space entanglement

Balasubramanian, V., McDermott, M. B., & Van Raamsdonk, M. Momentum-space entanglement and renormalization in quantum field theory. PRD 86(4), 045014 (2012).

Kunkel, P. et al. Detecting entanglement structure in continuous quantum many-body system. PRL 128, 020402 (2022)

* Field-space entanglement

$$\hat{H} = \hat{H}_A[\hat{\phi}_A(x)] + \hat{H}_B[\phi_B(x)] + \hat{H}_{\text{coupling}}[\hat{\phi}_A(x), \hat{\phi}_B(x)]$$

Mollabashi, A., Shiba, N., & Takayanagi, T. Entanglement between two interacting CFTs and generalized holographic entanglement entropy. JHEP 2014(4), 1-36 (2014).

Has not been detected in experiment!

Field-space Entanglement in Quantum Field Simulators (1)

IOPscience



Journals

Books

Publishing Support



Quantum Science and Technology

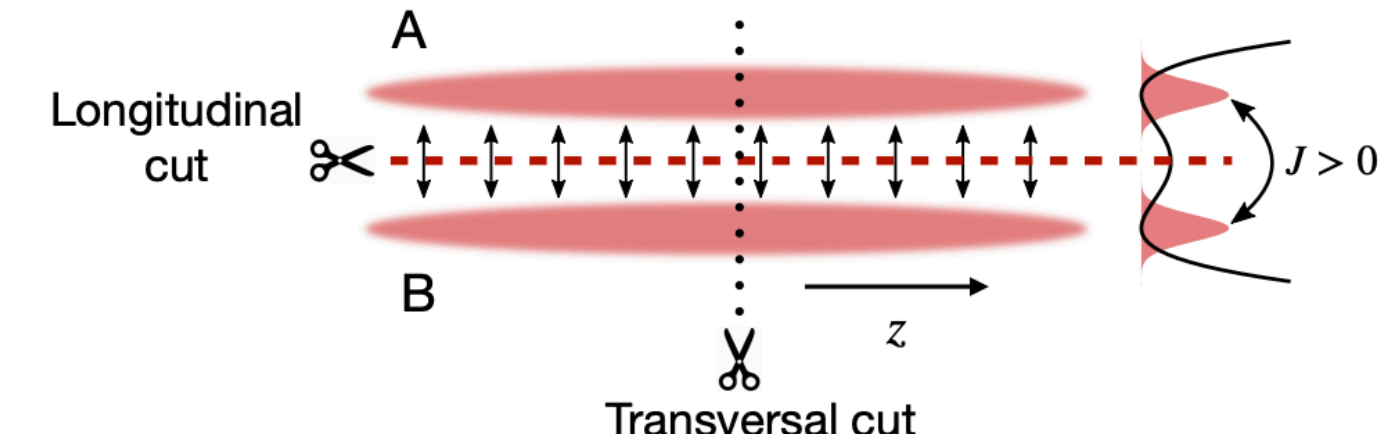
ACCEPTED MANUSCRIPT

Extensive entanglement between coupled Tomonaga-Luttinger liquids in and out of equilibrium

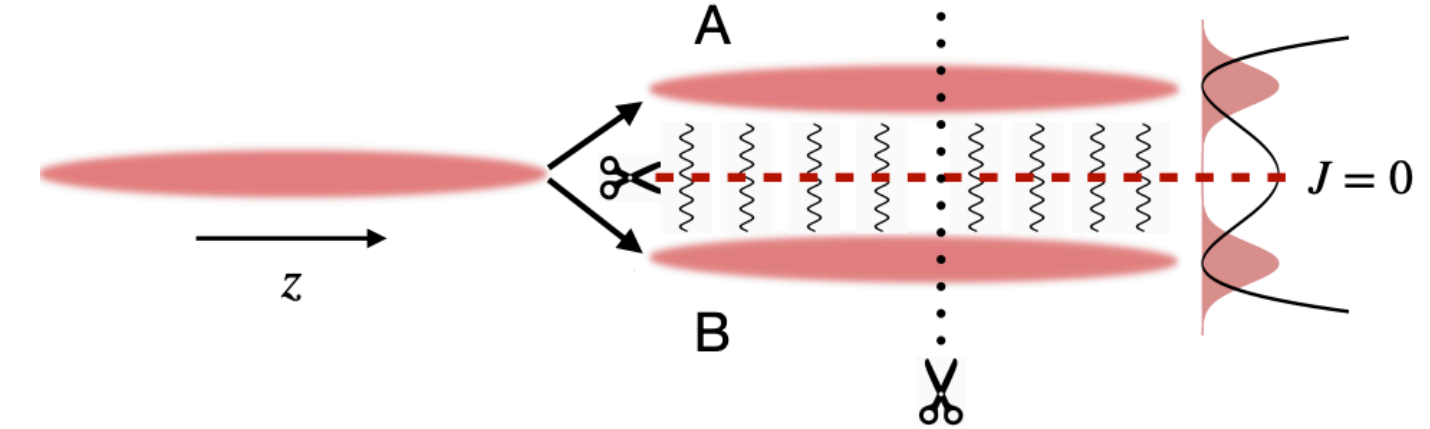
Taufiq Murtadho, Marek Gluza and Nelly Huei Ying Ng

Accepted Manuscript online 15 July 2026 • © 2026 IOP Publishing Ltd. All rights, including for text and data mining, AI training, and similar technologies, are reserved.

(a) Tunnel coupling



(b) Coherent splitting



Logarithmic Negativity $E_{\mathcal{N}}$

Mutual Information $I(A : B)$

Tunnel coupling

$$E_{\mathcal{N}} = \sum_{k>0}^{\Lambda} \max \left\{ 0, -\log \left[\sqrt{C_{k,J}(1 + 2\eta_k^+)(1 + 2\eta_k^-)} \right] \right\}$$

$$I(A : B) = \sum_{k>0}^{k_{\Lambda}} 2 \left[\left(\lambda_{k,J} + \frac{1}{2} \right) \log \left(\lambda_{k,J} + \frac{1}{2} \right) - \left(\lambda_{k,J} - \frac{1}{2} \right) \log \left(\lambda_{k,J} - \frac{1}{2} \right) \right] \\ - \sum_{a=\pm} \sum_{k>0}^{k_{\Lambda}} [(\eta_k^a + 1) \log (\eta_k^a + 1) - \eta_k^a \log \eta_k^a], \quad (25)$$

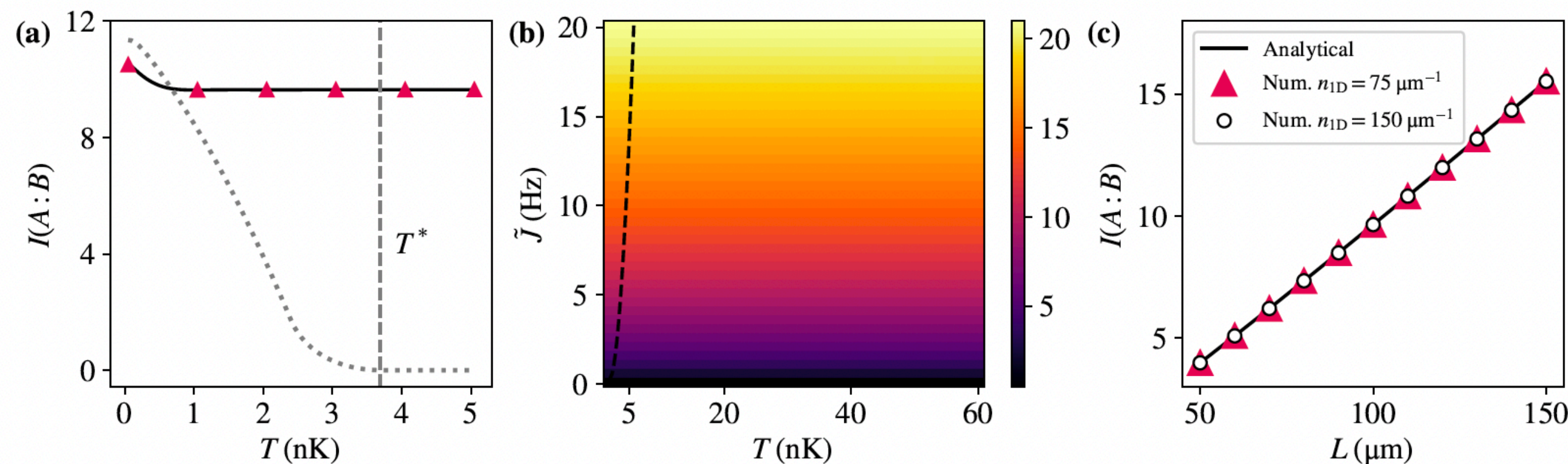
Coh. Splitting

$$E_{\mathcal{N}} = \sum_{a=\pm 1} \sum_{k>0}^{\Lambda} \max \left\{ 0, -\log \left[\sqrt{C_{k,r}^a(1 + 2\eta_k^+)} \right] \right\}$$

$$I(A : B) = \sum_{k>0}^{k_{\Lambda}} 2 \left[\left(\lambda_{k,r} + \frac{1}{2} \right) \log \left(\lambda_{k,r} + \frac{1}{2} \right) - \left(\lambda_{k,r} - \frac{1}{2} \right) \log \left(\lambda_{k,r} - \frac{1}{2} \right) \right] \\ - (1 + \eta_k^+) \log(1 + \eta_k^+) + \eta_k^+ \log \eta_k^+, \quad (31)$$

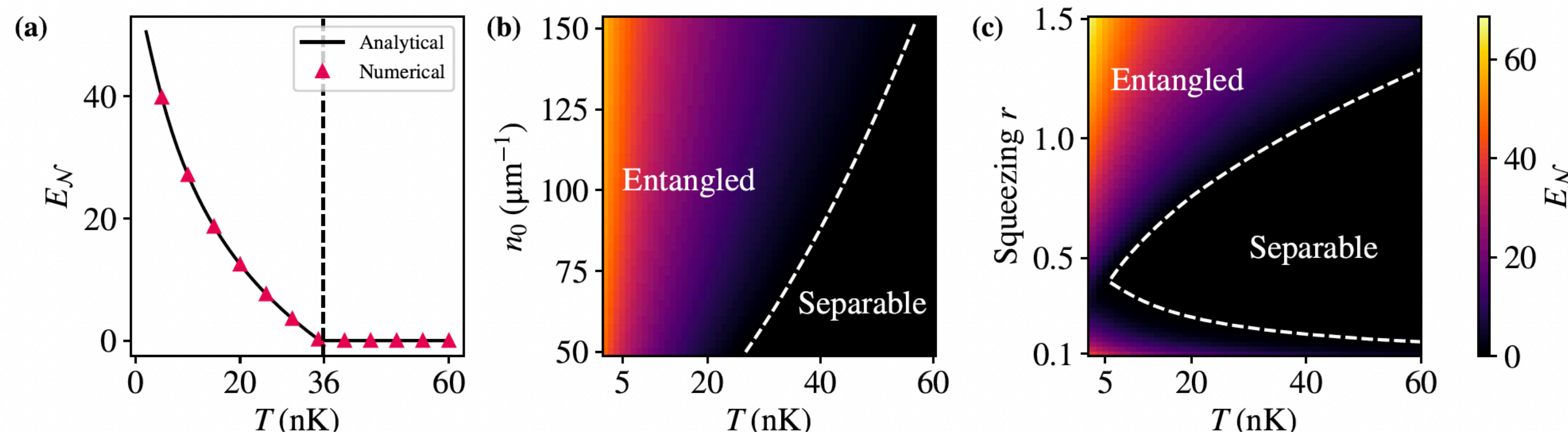
How “cold” is too “hot” for entanglement?

Tunnel-coupled



- * Entanglement drops, MI constant $\rightarrow$ **Quantum corr. transmutes into classical corr.**
- * **Entanglement/MI scales extensively with length, but not with density**

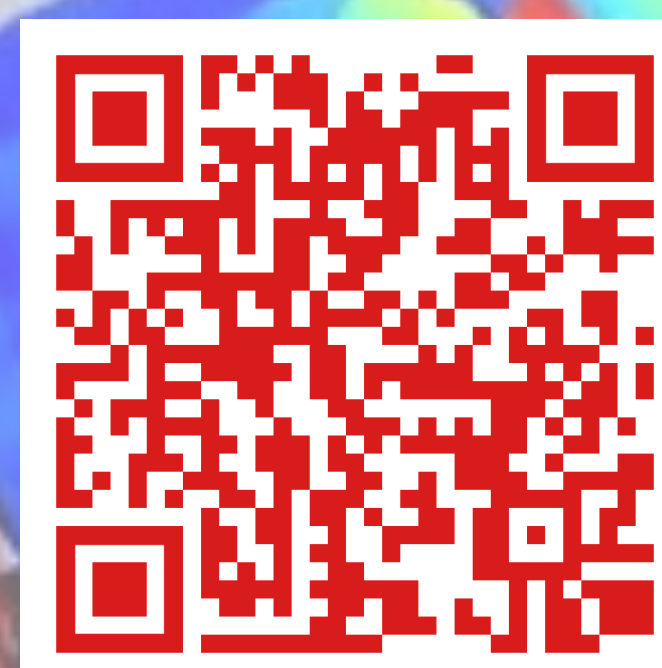
Coherently-split



$$T^* = \sup_{0 < k \leq k_\Lambda} \left\{ T_k^* = \frac{\varepsilon_k^+}{k_B} \left(\ln \left[\sqrt{\frac{C_{k,r} + C_{k,r}^{-1} + 2}{C_{k,r} + C_{k,r}^{-1} - 2}} \right] \right)^{-1} \right\}.$$

Entanglement can be detected at temperatures accessible in the current experiments!

Conclusion & Outlook



Take-home messages

- * **Common phase extraction** in ultracold Bose gases is **possible** to be done in experiments
- * **Reveals new frontiers in theoretical and experimental 1D quantum field simulators** by expanding the range of measurable quantities
- * Key application: **detecting entanglement *between* excitation of multiple fields (field-space entanglement)** at finite temperatures



Outlook

- * **Improve the accuracy** of common phase readout + full tomography
- * **Detect entanglement in experiments!**
- * **Implement in other systems** (e.g. monolayer and bilayer 2D Bose gases)
- * Apply to other use cases: **thermalization, non-Gaussianity,**



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